tropical: Knapsack

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1 Key results

This report covers 240 runs (5 seeds; 0 diverged) under frozen hash `057e46d824d4`.

Where to look. The headline metric is `ood_len_f1` (higher is better). Figure 1 plots it against `fishr_lambda`. Table 1 gives every cell's mean with its 95% bootstrap CI. Table 2 compares each level with the control. Table 3 gives the trend. Table 4 gives the paired contrasts. Its tests take the parametric path: values away from the metric bounds. Cells are compared within a seed: a seed index fixes the data and the initialization. Comparisons are marked significant at $p < 0.05$ after the stated correction.

1.1 `ood_len_f1`

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.6131	[0.598, 0.629]	0.0198	5	0
adaptive	pointer	0.01	0.6135	[0.594, 0.633]	0.0252	5	0
adaptive	pointer	0.1	0.6135	[0.595, 0.632]	0.0244	5	0
adaptive	pointer	1	0.6152	[0.596, 0.634]	0.0241	5	0
adaptive	pointer	10	0.6135	[0.595, 0.632]	0.0246	5	0
adaptive	pointer	100	0.6153	[0.597, 0.634]	0.0234	5	0
adaptive	pointer	1000	0.6145	[0.595, 0.634]	0.025	5	0
adaptive	pointer	1e+04	0.6178	[0.599, 0.637]	0.024	5	0
adaptive	seq2seq	0	0.5818	[0.573, 0.591]	0.0127	5	0
adaptive	seq2seq	0.01	0.5841	[0.574, 0.595]	0.0143	5	0
adaptive	seq2seq	0.1	0.584	[0.574, 0.594]	0.013	5	0
adaptive	seq2seq	1	0.5844	[0.574, 0.595]	0.0138	5	0
adaptive	seq2seq	10	0.585	[0.575, 0.599]	0.0152	5	0
adaptive	seq2seq	100	0.5828	[0.577, 0.59]	0.00899	5	0
adaptive	seq2seq	1000	0.5803	[0.576, 0.584]	0.00514	5	0
adaptive	seq2seq	1e+04	0.5748	[0.564, 0.586]	0.014	5	0
tropical	pointer	0	0.6177	[0.606, 0.629]	0.0158	5	0
tropical	pointer	0.01	0.6232	[0.607, 0.64]	0.0219	5	0
tropical	pointer	0.1	0.6167	[0.601, 0.634]	0.0215	5	0
tropical	pointer	1	0.6164	[0.604, 0.628]	0.0155	5	0
tropical	pointer	10	0.6151	[0.597, 0.637]	0.0262	5	0
tropical	pointer	100	0.6211	[0.602, 0.64]	0.0253	5	0
tropical	pointer	1000	0.6193	[0.607, 0.633]	0.016	5	0
tropical	pointer	1e+04	0.6165	[0.6, 0.633]	0.0212	5	0
tropical	seq2seq	0	0.6866	[0.671, 0.708]	0.025	5	0
tropical	seq2seq	0.01	0.6873	[0.672, 0.711]	0.0256	5	0
tropical	seq2seq	0.1	0.687	[0.672, 0.712]	0.0272	5	0
tropical	seq2seq	1	0.6834	[0.667, 0.705]	0.0251	5	0
tropical	seq2seq	10	0.6852	[0.672, 0.701]	0.0197	5	0
tropical	seq2seq	100	0.6891	[0.679, 0.701]	0.0133	5	0

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	1000	0.6903	[0.67, 0.711]	0.0264	5	0
tropical	seq2seq	1e+04	0.6486	[0.636, 0.669]	0.0219	5	0
vanilla	pointer	0	0.6038	[0.583, 0.624]	0.0273	5	0
vanilla	pointer	0.01	0.6111	[0.592, 0.63]	0.0247	5	0
vanilla	pointer	0.1	0.6111	[0.591, 0.631]	0.0255	5	0
vanilla	pointer	1	0.611	[0.591, 0.631]	0.0259	5	0
vanilla	pointer	10	0.6103	[0.591, 0.629]	0.0246	5	0
vanilla	pointer	100	0.6112	[0.592, 0.631]	0.0252	5	0
vanilla	pointer	1000	0.6133	[0.594, 0.632]	0.0243	5	0
vanilla	pointer	1e+04	0.6194	[0.599, 0.638]	0.024	5	0
vanilla	seq2seq	0	0.5878	[0.576, 0.608]	0.0221	5	0
vanilla	seq2seq	0.01	0.5978	[0.579, 0.619]	0.0255	5	0
vanilla	seq2seq	0.1	0.5972	[0.577, 0.62]	0.0273	5	0
vanilla	seq2seq	1	0.5964	[0.578, 0.616]	0.0246	5	0
vanilla	seq2seq	10	0.5974	[0.579, 0.621]	0.0264	5	0
vanilla	seq2seq	100	0.6016	[0.582, 0.624]	0.0264	5	0
vanilla	seq2seq	1000	0.6102	[0.583, 0.637]	0.0365	5	0
vanilla	seq2seq	1e+04	0.5856	[0.574, 0.604]	0.0203	5	0

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	4.704×10^{-4}	[-0.0412, 0.0421]	1.0000
adaptive	pointer	0.1	4.955×10^{-4}	[-0.0412, 0.0422]	1.0000
adaptive	pointer	1	0.002111	[-0.0396, 0.0438]	1.0000
adaptive	pointer	10	4.038×10^{-4}	[-0.0413, 0.0421]	1.0000
adaptive	pointer	100	0.002215	[-0.0395, 0.0439]	1.0000
adaptive	pointer	1000	0.00142	[-0.0402, 0.0431]	1.0000
adaptive	pointer	1e+04	0.004714	[-0.037, 0.0464]	0.9998
adaptive	seq2seq	0.01	0.0023	[-0.0196, 0.0242]	0.9998
adaptive	seq2seq	0.1	0.002296	[-0.0196, 0.0242]	0.9999
adaptive	seq2seq	1	0.00262	[-0.0193, 0.0245]	0.9996
adaptive	seq2seq	10	0.003213	[-0.0187, 0.0251]	0.9987
adaptive	seq2seq	100	0.001079	[-0.0208, 0.023]	1.0000
adaptive	seq2seq	1000	-0.001416	[-0.0233, 0.0205]	1.0000
adaptive	seq2seq	1e+04	-0.006909	[-0.0288, 0.015]	0.9169
tropical	pointer	0.01	0.005519	[-0.0308, 0.0418]	0.9984
tropical	pointer	0.1	-0.001068	[-0.0374, 0.0353]	1.0000
tropical	pointer	1	-0.001279	[-0.0376, 0.035]	1.0000
tropical	pointer	10	-0.002667	[-0.039, 0.0337]	1.0000
tropical	pointer	100	0.003386	[-0.0329, 0.0397]	0.9999
tropical	pointer	1000	0.001607	[-0.0347, 0.0379]	1.0000
tropical	pointer	1e+04	-0.001233	[-0.0376, 0.0351]	1.0000
tropical	seq2seq	0.01	7.157×10^{-4}	[-0.0402, 0.0416]	1.0000
tropical	seq2seq	0.1	3.535×10^{-4}	[-0.0406, 0.0413]	1.0000
tropical	seq2seq	1	-0.003218	[-0.0442, 0.0377]	1.0000
tropical	seq2seq	10	-0.001459	[-0.0424, 0.0395]	1.0000
tropical	seq2seq	100	0.0025	[-0.0384, 0.0434]	1.0000
tropical	seq2seq	1000	0.003661	[-0.0373, 0.0446]	0.9999
tropical	seq2seq	1e+04	-0.03806	[-0.079, 0.00288]	0.0770
vanilla	pointer	0.01	0.007255	[-0.0368, 0.0513]	0.9974
vanilla	pointer	0.1	0.007237	[-0.0368, 0.0513]	0.9974
vanilla	pointer	1	0.007177	[-0.0369, 0.0512]	0.9975

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
vanilla	pointer	10	0.006489	[-0.0376, 0.0505]	0.9987
vanilla	pointer	100	0.007339	[-0.0367, 0.0514]	0.9972
vanilla	pointer	1000	0.009424	[-0.0346, 0.0535]	0.9879
vanilla	pointer	1e+04	0.01554	[-0.0285, 0.0596]	0.8684
vanilla	seq2seq	0.01	0.01002	[-0.0363, 0.0564]	0.9872
vanilla	seq2seq	0.1	0.009458	[-0.0369, 0.0558]	0.9907
vanilla	seq2seq	1	0.008603	[-0.0377, 0.055]	0.9946
vanilla	seq2seq	10	0.00966	[-0.0367, 0.056]	0.9896
vanilla	seq2seq	100	0.01379	[-0.0326, 0.0601]	0.9358
vanilla	seq2seq	1000	0.02244	[-0.0239, 0.0688]	0.6321
vanilla	seq2seq	1e+04	-0.002174	[-0.0485, 0.0442]	1.0000

Table 3: `ood_len_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	5.244×10^{-4}	$[-3.53 \times 10^{-5}, 0.00148]$	0.7842	0.0514	0.7075
adaptive	seq2seq	-0.001307	[-0.00387, 0.00102]	0.1962	-0.0629	0.6395
tropical	pointer	-3.658×10^{-4}	$[-9.31 \times 10^{-4}, 3.80 \times 10^{-4}]$	0.8293	-0.00571	0.9794
tropical	seq2seq	-0.003714	[-0.00604, -0.00124]	0.0808	-0.143	0.2897
vanilla	pointer	0.00105	$[3.87 \times 10^{-4}, 0.00189]$	0.5936	0.109	0.4181
vanilla	seq2seq	-1.934×10^{-4}	$[-5.01 \times 10^{-4}, 1.13 \times 10^{-4}]$	0.9302	0.0362	0.7964

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.01387	[-0.0164, 0.0441]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.01214	[-0.00917, 0.0334]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.005568	[-0.0228, 0.0339]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.005418	[-0.0188, 0.0297]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.004717	[-0.0232, 0.0327]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.00992	[-0.018, 0.0379]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.006056	[-0.0139, 0.026]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-0.002901	[-0.0255, 0.0197]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.09885	[0.0491, 0.149]	0.0790
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.08955	[0.0369, 0.142]	0.1282
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.08975	[0.0337, 0.146]	0.1351
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.08703	[0.0284, 0.146]	0.1511

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.08773	[0.0358, 0.14]	0.1282
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.08756	[0.0447, 0.13]	0.0761
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.08007	[0.0134, 0.147]	0.2611
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.06297	[0.0213, 0.105]	0.1511
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.009205	[-0.00937, 0.0278]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.00242	$[-3.62 \times 10^{-4}, 0.0052]$	0.9509
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.002463	$[-7.97 \times 10^{-4}, 0.00572]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.004139	[0.00103, 0.00725]	0.3133
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.003119	$[3.15 \times 10^{-4}, 0.00592]$	0.5129
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.004081	[0.0017, 0.00646]	0.1417
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.0012	[-0.00371, 0.00611]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.001622	[-0.00446, 0.00121]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.006026	[-0.0262, 0.0141]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.01374	[-0.0423, 0.0148]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.01319	[-0.0443, 0.0179]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.01201	[-0.0379, 0.0139]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.01247	[-0.035, 0.01]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.01874	[-0.0449, 0.00742]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.02989	[-0.0762, 0.0165]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.01076	[-0.046, 0.0244]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.0313	[0.00424, 0.0584]	0.4884
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.02947	[-0.00502, 0.064]	0.7317
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.0295	[-0.00418, 0.0632]	0.7317
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.03079	[-0.00336, 0.0649]	0.7317
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.02849	[-0.00317, 0.0602]	0.7317
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.03244	[0.005, 0.0599]	0.4869
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.03414	[-0.00227, 0.0705]	0.7181
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.04293	[0.00424, 0.0816]	0.5167

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.06891	[-0.0986, -0.0392]	0.0654	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.0641	[-0.0981, -0.0301]	0.1211	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.07033	[-0.104, -0.0366]	0.0888	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.06697	[-0.0971, -0.0369]	0.0733	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.07012	[-0.0974, -0.0429]	0.0487	*
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.06802	[-0.0952, -0.0409]	0.0517	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.07096	[-0.118, -0.0237]	0.2536	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.03208	[-0.0621, -0.00204]	0.5375	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.01607	[-0.00908, 0.0412]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.01331	[-0.0219, 0.0485]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.01385	[-0.023, 0.0507]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.01465	[-0.0225, 0.0518]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.0129	[-0.0211, 0.0469]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.009619	[-0.0298, 0.0491]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.003052	[-0.0492, 0.0553]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.03379	[0.00843, 0.0591]	0.3545	

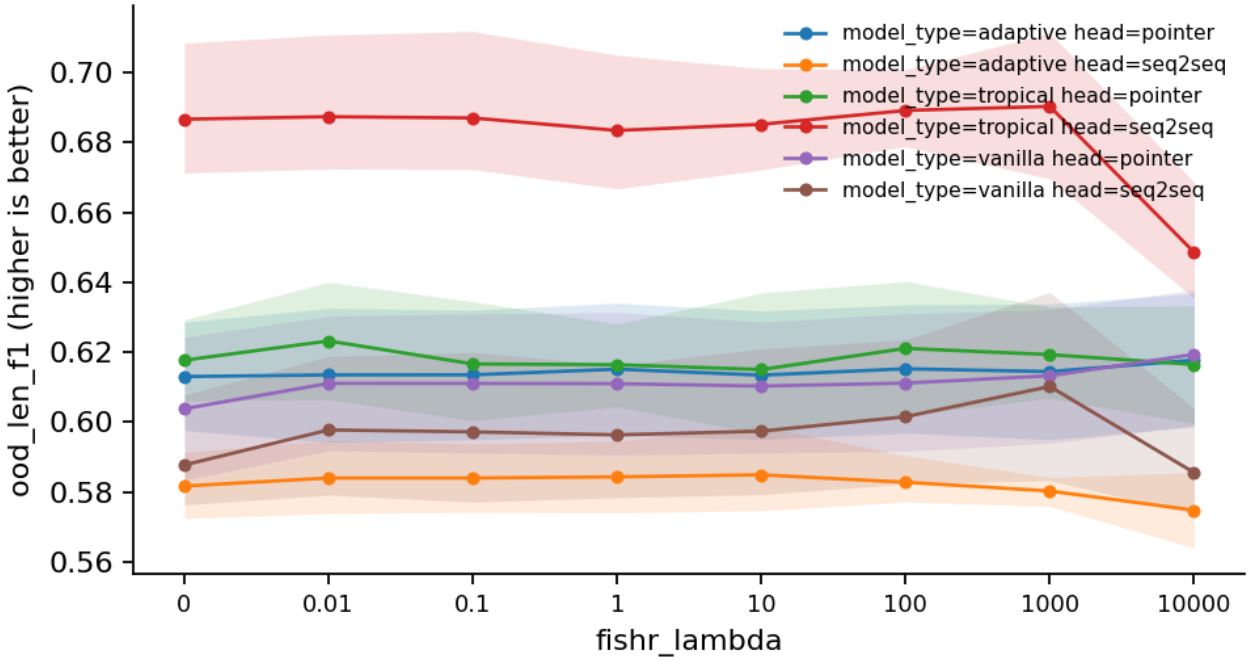


Figure 1: `ood_len_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

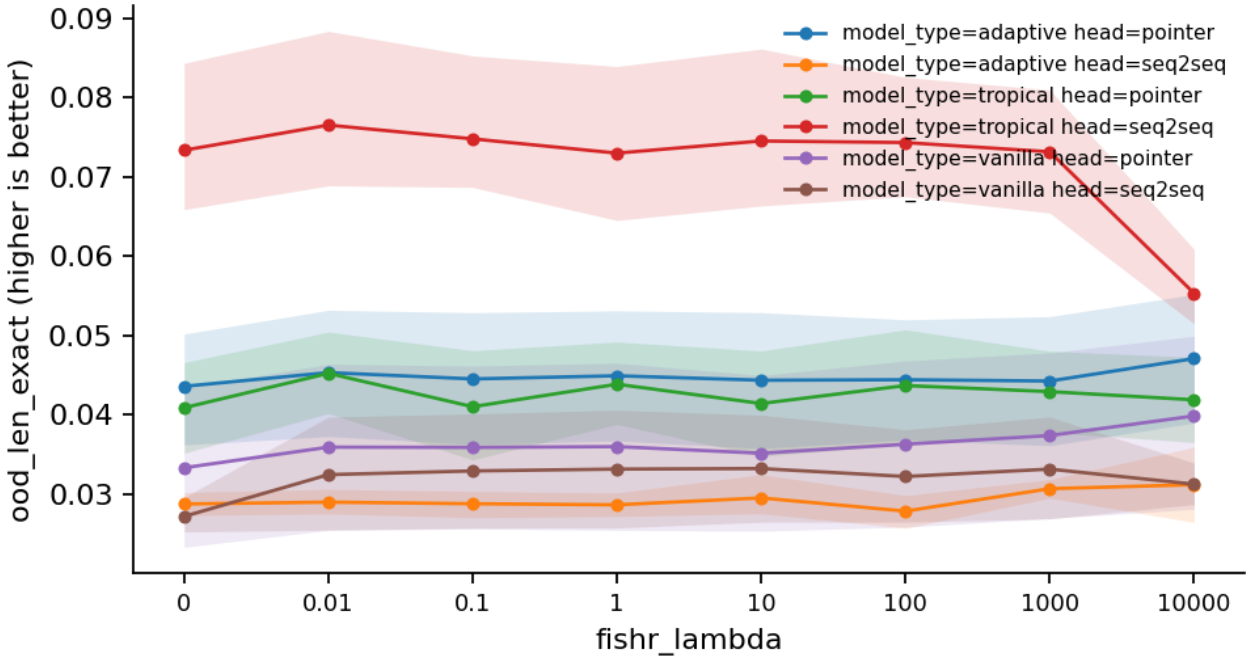


Figure 2: `ood_len_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A Other metrics

A.1 `ood_len_exact`

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.0436	[0.0362, 0.0502]	0.00887	5	0
adaptive	pointer	0.01	0.04533	[0.0373, 0.0532]	0.0105	5	0
adaptive	pointer	0.1	0.04453	[0.0363, 0.0529]	0.0106	5	0
adaptive	pointer	1	0.04493	[0.0367, 0.0531]	0.0107	5	0
adaptive	pointer	10	0.04437	[0.0358, 0.0529]	0.0109	5	0
adaptive	pointer	100	0.04443	[0.0369, 0.052]	0.00975	5	0
adaptive	pointer	1000	0.04427	[0.0361, 0.0524]	0.0104	5	0
adaptive	pointer	1e+04	0.0471	[0.039, 0.0552]	0.0103	5	0
adaptive	seq2seq	0	0.02877	[0.0273, 0.0302]	0.00187	5	0
adaptive	seq2seq	0.01	0.029	[0.0275, 0.0306]	0.00197	5	0
adaptive	seq2seq	0.1	0.0288	[0.0271, 0.0304]	0.00213	5	0
adaptive	seq2seq	1	0.02867	[0.0272, 0.0302]	0.00193	5	0
adaptive	seq2seq	10	0.02953	[0.0276, 0.0325]	0.00324	5	0
adaptive	seq2seq	100	0.02787	[0.0257, 0.0298]	0.00259	5	0
adaptive	seq2seq	1000	0.0307	[0.0295, 0.0319]	0.0015	5	0
adaptive	seq2seq	1e+04	0.0312	[0.0265, 0.0359]	0.00615	5	0
tropical	pointer	0	0.0409	[0.0352, 0.0466]	0.00734	5	0
tropical	pointer	0.01	0.04523	[0.0402, 0.0504]	0.00673	5	0
tropical	pointer	0.1	0.04103	[0.0344, 0.0481]	0.00885	5	0
tropical	pointer	1	0.04387	[0.0388, 0.0492]	0.00661	5	0
tropical	pointer	10	0.04143	[0.0348, 0.0481]	0.00833	5	0
tropical	pointer	100	0.0437	[0.0363, 0.0507]	0.00918	5	0

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	pointer	1000	0.04293	[0.0378, 0.048]	0.00669	5	0
tropical	pointer	1e+04	0.0419	[0.0365, 0.0473]	0.00729	5	0
tropical	seq2seq	0	0.07337	[0.0659, 0.0843]	0.0124	5	0
tropical	seq2seq	0.01	0.07653	[0.0689, 0.0884]	0.0135	5	0
tropical	seq2seq	0.1	0.0748	[0.0687, 0.0853]	0.0118	5	0
tropical	seq2seq	1	0.073	[0.0645, 0.0839]	0.0131	5	0
tropical	seq2seq	10	0.07453	[0.0664, 0.0861]	0.0129	5	0
tropical	seq2seq	100	0.07433	[0.0675, 0.0826]	0.00937	5	0
tropical	seq2seq	1000	0.07317	[0.0655, 0.081]	0.0103	5	0
tropical	seq2seq	1e+04	0.05533	[0.0515, 0.0609]	0.00623	5	0
vanilla	pointer	0	0.03333	[0.0233, 0.0434]	0.0135	5	0
vanilla	pointer	0.01	0.03593	[0.0255, 0.0464]	0.0135	5	0
vanilla	pointer	0.1	0.0359	[0.0256, 0.0462]	0.0134	5	0
vanilla	pointer	1	0.036	[0.0255, 0.0465]	0.0137	5	0
vanilla	pointer	10	0.03517	[0.0254, 0.045]	0.0131	5	0
vanilla	pointer	100	0.0363	[0.0259, 0.0468]	0.0134	5	0
vanilla	pointer	1000	0.0374	[0.027, 0.0478]	0.0136	5	0
vanilla	pointer	1e+04	0.03987	[0.0281, 0.0499]	0.0138	5	0
vanilla	seq2seq	0	0.0272	[0.0253, 0.0296]	0.00273	5	0
vanilla	seq2seq	0.01	0.03247	[0.0255, 0.0397]	0.00919	5	0
vanilla	seq2seq	0.1	0.03293	[0.0257, 0.0401]	0.00946	5	0
vanilla	seq2seq	1	0.03317	[0.0258, 0.0406]	0.00968	5	0
vanilla	seq2seq	10	0.03323	[0.0265, 0.04]	0.0087	5	0
vanilla	seq2seq	100	0.03223	[0.0265, 0.0381]	0.00749	5	0
vanilla	seq2seq	1000	0.03317	[0.0269, 0.0397]	0.0083	5	0
vanilla	seq2seq	1e+04	0.03127	[0.0287, 0.0339]	0.00367	5	0

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0.001733	[-0.0162, 0.0197]	0.9999
adaptive	pointer	0.1	9.333×10^{-4}	[-0.017, 0.0189]	1.0000
adaptive	pointer	1	0.001333	[-0.0166, 0.0193]	1.0000
adaptive	pointer	10	7.667×10^{-4}	[-0.0172, 0.0187]	1.0000
adaptive	pointer	100	8.333×10^{-4}	[-0.0171, 0.0188]	1.0000
adaptive	pointer	1000	6.667×10^{-4}	[-0.0173, 0.0186]	1.0000
adaptive	pointer	1e+04	0.0035	[-0.0144, 0.0214]	0.9928
adaptive	seq2seq	0.01	2.333×10^{-4}	[-0.00504, 0.00551]	1.0000
adaptive	seq2seq	0.1	3.333×10^{-5}	[-0.00524, 0.00531]	1.0000
adaptive	seq2seq	1	-1.000×10^{-4}	[-0.00537, 0.00517]	1.0000
adaptive	seq2seq	10	7.667×10^{-4}	[-0.00451, 0.00604]	0.9988
adaptive	seq2seq	100	-9.000×10^{-4}	[-0.00617, 0.00437]	0.9967
adaptive	seq2seq	1000	0.001933	[-0.00334, 0.00721]	0.8474
adaptive	seq2seq	1e+04	0.002433	[-0.00284, 0.00771]	0.6768
tropical	pointer	0.01	0.004333	[-0.00909, 0.0178]	0.9082
tropical	pointer	0.1	1.333×10^{-4}	[-0.0133, 0.0136]	1.0000
tropical	pointer	1	0.002967	[-0.0105, 0.0164]	0.9856
tropical	pointer	10	5.333×10^{-4}	[-0.0129, 0.014]	1.0000
tropical	pointer	100	0.0028	[-0.0106, 0.0162]	0.9895
tropical	pointer	1000	0.002033	[-0.0114, 0.0155]	0.9984
tropical	pointer	1e+04	0.001	[-0.0124, 0.0144]	1.0000
tropical	seq2seq	0.01	0.003167	[-0.0168, 0.0231]	0.9979
tropical	seq2seq	0.1	0.001433	[-0.0185, 0.0214]	1.0000

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
tropical	seq2seq	1	-3.667×10^{-4}	[-0.0203, 0.0196]	1.0000
tropical	seq2seq	10	0.001167	[-0.0188, 0.0211]	1.0000
tropical	seq2seq	100	9.667×10^{-4}	[-0.019, 0.0209]	1.0000
tropical	seq2seq	1000	-2.000×10^{-4}	[-0.0202, 0.0198]	1.0000
tropical	seq2seq	1e+04	-0.01803	[-0.038, 0.00195]	0.0904
vanilla	pointer	0.01	0.0026	[-0.021, 0.0262]	0.9998
vanilla	pointer	0.1	0.002567	[-0.021, 0.0262]	0.9998
vanilla	pointer	1	0.002667	[-0.0209, 0.0263]	0.9998
vanilla	pointer	10	0.001833	[-0.0218, 0.0254]	1.0000
vanilla	pointer	100	0.002967	[-0.0206, 0.0266]	0.9995
vanilla	pointer	1000	0.004067	[-0.0195, 0.0277]	0.9965
vanilla	pointer	1e+04	0.006533	[-0.0171, 0.0301]	0.9540
vanilla	seq2seq	0.01	0.005267	[-0.00839, 0.0189]	0.8167
vanilla	seq2seq	0.1	0.005733	[-0.00793, 0.0194]	0.7565
vanilla	seq2seq	1	0.005967	[-0.00769, 0.0196]	0.7243
vanilla	seq2seq	10	0.006033	[-0.00763, 0.0197]	0.7150
vanilla	seq2seq	100	0.005033	[-0.00863, 0.0187]	0.8445
vanilla	seq2seq	1000	0.005967	[-0.00769, 0.0196]	0.7243
vanilla	seq2seq	1e+04	0.004067	[-0.00959, 0.0177]	0.9357

Table 7: `ood_len_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	1.524×10^{-4}	$[-2.26 \times 10^{-4}, 5.17 \times 10^{-4}]$	0.8531	-0.0133	0.9224
adaptive	seq2seq	3.429×10^{-4}	$[-2.38 \times 10^{-4}, 8.18 \times 10^{-4}]$	0.1898	0.0705	0.6169
tropical	pointer	-2.274×10^{-4}	$[-5.62 \times 10^{-4}, 1.67 \times 10^{-4}]$	0.7131	-0.0552	0.6689
tropical	seq2seq	-0.00234	[-0.0036, -0.0013]	0.0223	-0.278	0.0410
vanilla	pointer	5.393×10^{-4}	$[2.43 \times 10^{-4}, 9.45 \times 10^{-4}]$	0.6126	0.0971	0.4719
vanilla	seq2seq	-1.452×10^{-4}	$[-8.81 \times 10^{-4}, 4.36 \times 10^{-4}]$	0.8240	0.061	0.6559

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.007567	[-0.0133, 0.0284]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.0093	[-0.00944, 0.028]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.005133	[-0.0171, 0.0273]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.007867	[-0.0139, 0.0296]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.006267	[-0.0149, 0.0275]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.0074	[-0.0137, 0.0285]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.005533	[-0.0139, 0.0249]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.002033	[-0.0204, 0.0245]	1.0000

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.04617	[0.0291, 0.0632]	0.0268	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.04407	[0.0191, 0.069]	0.0886	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.04187	[0.0196, 0.0642]	0.0840	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.03983	[0.013, 0.0667]	0.1318	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.0413	[0.0171, 0.0655]	0.0900	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.0421	[0.0232, 0.061]	0.0486	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.04	[0.0186, 0.0614]	0.0840	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.02407	[0.014, 0.0341]	0.0400	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.01027	[-0.00237, 0.0229]	0.7845	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.0094	[0.0045, 0.0143]	0.0956	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.008633	[0.00327, 0.014]	0.1660	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.008933	[0.00245, 0.0154]	0.2246	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.0092	[0.00286, 0.0155]	0.2048	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.008133	[0.00279, 0.0135]	0.1882	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.006867	$[6.30 \times 10^{-4}, 0.0131]$	0.3777	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.007233	[0.00112, 0.0133]	0.3331	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.001567	$[-8.49 \times 10^{-4}, 0.00398]$	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.003467	[-0.0143, 0.00734]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.004133	[-0.0155, 0.00725]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.0045	[-0.016, 0.007]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.0037	[-0.0137, 0.00631]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.004367	[-0.0135, 0.0048]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.002467	[-0.0136, 0.0087]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-6.667×10^{-5}	[-0.011, 0.0108]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.01483	[0.00448, 0.0252]	0.2555	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.01633	[0.00503, 0.0276]	0.2555	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.01573	[0.00425, 0.0272]	0.2664	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.01627	[0.00413, 0.0284]	0.2664	

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.01483	[0.00349, 0.0262]	0.2664
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.01657	[0.00673, 0.0264]	0.1992
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.01357	$[-8.19 \times 10^{-4}, 0.028]$	0.5810
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.0159	$[8.74 \times 10^{-4}, 0.0309]$	0.4673
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.03247	[-0.0531, -0.0118]	0.2277
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.0313	[-0.0515, -0.0111]	0.2277
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.03377	[-0.0544, -0.0131]	0.2093
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.02913	[-0.0479, -0.0103]	0.2277
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.0331	[-0.0522, -0.014]	0.1904
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.03063	[-0.0471, -0.0142]	0.1607
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.03023	[-0.0469, -0.0136]	0.1682
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.01343	$[-0.0276, 7.40 \times 10^{-4}]$	0.5810
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.006133	[-0.00832, 0.0206]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.003467	[-0.0159, 0.0228]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.002967	[-0.0158, 0.0217]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.002833	[-0.0165, 0.0222]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.001933	[-0.0155, 0.0193]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.004067	[-0.0148, 0.023]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.004233	[-0.015, 0.0234]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.0086	[-0.00717, 0.0244]	1.0000

A.2 `ood_val_f1`

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.8498	[0.833, 0.861]	0.0172	5	0
adaptive	pointer	0.01	0.8559	[0.851, 0.859]	0.00539	5	0
adaptive	pointer	0.1	0.8534	[0.848, 0.859]	0.00739	5	0
adaptive	pointer	1	0.8563	[0.851, 0.861]	0.0062	5	0
adaptive	pointer	10	0.8555	[0.85, 0.861]	0.00677	5	0
adaptive	pointer	100	0.8537	[0.85, 0.858]	0.00514	5	0
adaptive	pointer	1000	0.8572	[0.851, 0.863]	0.00768	5	0
adaptive	pointer	1e+04	0.8536	[0.845, 0.862]	0.0115	5	0

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	seq2seq	0	0.8555	[0.85, 0.86]	0.00636	5	0
adaptive	seq2seq	0.01	0.8505	[0.842, 0.857]	0.0101	5	0
adaptive	seq2seq	0.1	0.8525	[0.845, 0.858]	0.00829	5	0
adaptive	seq2seq	1	0.8474	[0.839, 0.856]	0.0118	5	0
adaptive	seq2seq	10	0.8424	[0.826, 0.856]	0.0202	5	0
adaptive	seq2seq	100	0.849	[0.84, 0.855]	0.00972	5	0
adaptive	seq2seq	1000	0.8472	[0.845, 0.849]	0.0024	5	0
adaptive	seq2seq	1e+04	0.8533	[0.848, 0.859]	0.00696	5	0
tropical	pointer	0	0.8171	[0.801, 0.836]	0.0222	5	0
tropical	pointer	0.01	0.8112	[0.79, 0.83]	0.0261	5	0
tropical	pointer	0.1	0.8137	[0.793, 0.834]	0.026	5	0
tropical	pointer	1	0.8133	[0.792, 0.832]	0.0254	5	0
tropical	pointer	10	0.8071	[0.782, 0.83]	0.0307	5	0
tropical	pointer	100	0.8046	[0.788, 0.824]	0.0236	5	0
tropical	pointer	1000	0.8166	[0.798, 0.83]	0.0209	5	0
tropical	pointer	1e+04	0.8028	[0.783, 0.825]	0.0285	5	0
tropical	seq2seq	0	0.7977	[0.784, 0.811]	0.0182	5	0
tropical	seq2seq	0.01	0.7866	[0.777, 0.801]	0.0159	5	0
tropical	seq2seq	0.1	0.7941	[0.785, 0.803]	0.0111	5	0
tropical	seq2seq	1	0.7971	[0.783, 0.816]	0.0221	5	0
tropical	seq2seq	10	0.7982	[0.783, 0.814]	0.0197	5	0
tropical	seq2seq	100	0.7928	[0.78, 0.806]	0.0175	5	0
tropical	seq2seq	1000	0.791	[0.776, 0.806]	0.0201	5	0
tropical	seq2seq	1e+04	0.7879	[0.776, 0.798]	0.0137	5	0
vanilla	pointer	0	0.851	[0.837, 0.862]	0.0166	5	0
vanilla	pointer	0.01	0.855	[0.85, 0.86]	0.0071	5	0
vanilla	pointer	0.1	0.8539	[0.848, 0.859]	0.00777	5	0
vanilla	pointer	1	0.8528	[0.845, 0.859]	0.00857	5	0
vanilla	pointer	10	0.854	[0.848, 0.86]	0.00852	5	0
vanilla	pointer	100	0.8522	[0.846, 0.859]	0.00868	5	0
vanilla	pointer	1000	0.8524	[0.846, 0.859]	0.00882	5	0
vanilla	pointer	1e+04	0.8576	[0.851, 0.862]	0.00779	5	0
vanilla	seq2seq	0	0.8501	[0.843, 0.856]	0.00851	5	0
vanilla	seq2seq	0.01	0.8546	[0.852, 0.857]	0.00284	5	0
vanilla	seq2seq	0.1	0.8557	[0.852, 0.859]	0.00434	5	0
vanilla	seq2seq	1	0.8555	[0.853, 0.858]	0.00326	5	0
vanilla	seq2seq	10	0.8545	[0.852, 0.857]	0.00338	5	0
vanilla	seq2seq	100	0.8473	[0.834, 0.857]	0.0154	5	0
vanilla	seq2seq	1000	0.8504	[0.839, 0.857]	0.0123	5	0
vanilla	seq2seq	1e+04	0.854	[0.85, 0.858]	0.0047	5	0

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0.006186	[-0.00995, 0.0223]	0.8205
adaptive	pointer	0.1	0.003626	[-0.0125, 0.0198]	0.9842
adaptive	pointer	1	0.006519	[-0.00962, 0.0227]	0.7850
adaptive	pointer	10	0.005776	[-0.0104, 0.0219]	0.8607
adaptive	pointer	100	0.003906	[-0.0122, 0.02]	0.9765
adaptive	pointer	1000	0.007405	[-0.00873, 0.0235]	0.6818
adaptive	pointer	1e+04	0.003867	[-0.0123, 0.02]	0.9778
adaptive	seq2seq	0.01	-0.005045	[-0.0237, 0.0136]	0.9585
adaptive	seq2seq	0.1	-0.002996	[-0.0216, 0.0156]	0.9977

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
adaptive	seq2seq	1	-0.008075	[-0.0267, 0.0105]	0.7298
adaptive	seq2seq	10	-0.01315	[-0.0318, 0.00546]	0.2583
adaptive	seq2seq	100	-0.006514	[-0.0251, 0.0121]	0.8722
adaptive	seq2seq	1000	-0.008361	[-0.027, 0.0102]	0.7004
adaptive	seq2seq	1e+04	-0.002203	[-0.0208, 0.0164]	0.9997
tropical	pointer	0.01	-0.00582	[-0.0506, 0.0389]	0.9994
tropical	pointer	0.1	-0.003398	[-0.0481, 0.0413]	1.0000
tropical	pointer	1	-0.0038	[-0.0485, 0.0409]	1.0000
tropical	pointer	10	-0.009917	[-0.0546, 0.0348]	0.9853
tropical	pointer	100	-0.01251	[-0.0572, 0.0322]	0.9518
tropical	pointer	1000	-5.000×10^{-4}	[-0.0452, 0.0442]	1.0000
tropical	pointer	1e+04	-0.01429	[-0.059, 0.0304]	0.9122
tropical	seq2seq	0.01	-0.01112	[-0.0419, 0.0197]	0.8557
tropical	seq2seq	0.1	-0.003676	[-0.0345, 0.0271]	0.9997
tropical	seq2seq	1	-6.708×10^{-4}	[-0.0315, 0.0301]	1.0000
tropical	seq2seq	10	4.043×10^{-4}	[-0.0304, 0.0312]	1.0000
tropical	seq2seq	100	-0.004937	[-0.0357, 0.0259]	0.9978
tropical	seq2seq	1000	-0.006704	[-0.0375, 0.0241]	0.9867
tropical	seq2seq	1e+04	-0.009891	[-0.0407, 0.0209]	0.9101
vanilla	pointer	0.01	0.004062	[-0.0128, 0.0209]	0.9771
vanilla	pointer	0.1	0.002917	[-0.0139, 0.0198]	0.9965
vanilla	pointer	1	0.00182	[-0.015, 0.0187]	0.9998
vanilla	pointer	10	0.002992	[-0.0139, 0.0199]	0.9959
vanilla	pointer	100	0.001215	[-0.0156, 0.0181]	1.0000
vanilla	pointer	1000	0.001418	[-0.0154, 0.0183]	1.0000
vanilla	pointer	1e+04	0.006583	[-0.0103, 0.0234]	0.8086
vanilla	seq2seq	0.01	0.004462	[-0.00978, 0.0187]	0.9191
vanilla	seq2seq	0.1	0.005581	[-0.00866, 0.0198]	0.8060
vanilla	seq2seq	1	0.005373	[-0.00887, 0.0196]	0.8303
vanilla	seq2seq	10	0.004395	[-0.00984, 0.0186]	0.9244
vanilla	seq2seq	100	-0.00281	[-0.017, 0.0114]	0.9923
vanilla	seq2seq	1000	2.845×10^{-4}	[-0.014, 0.0145]	1.0000
vanilla	seq2seq	1e+04	0.003896	[-0.0103, 0.0181]	0.9567

Table 11: `ood_val_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-7.179×10^{-5}	$[-7.42 \times 10^{-4}, 7.15 \times 10^{-4}]$	0.9039	-0.0019	1.0000
adaptive	seq2seq	-2.291×10^{-5}	$[-8.70 \times 10^{-4}, 9.73 \times 10^{-4}]$	0.9802	-0.0743	0.5815
tropical	pointer	-0.001012	$[-0.00314, 4.09 \times 10^{-4}]$	0.6271	-0.0819	0.5483
tropical	seq2seq	-2.368×10^{-4}	[-0.00151, 0.00148]	0.8676	-0.0438	0.7620
vanilla	pointer	1.414×10^{-4}	$[-7.56 \times 10^{-4}, 0.00101]$	0.8305	0.0514	0.7113
vanilla	seq2seq	-7.312×10^{-4}	$[-0.00195, 6.20 \times 10^{-5}]$	0.2806	-0.0819	0.5595

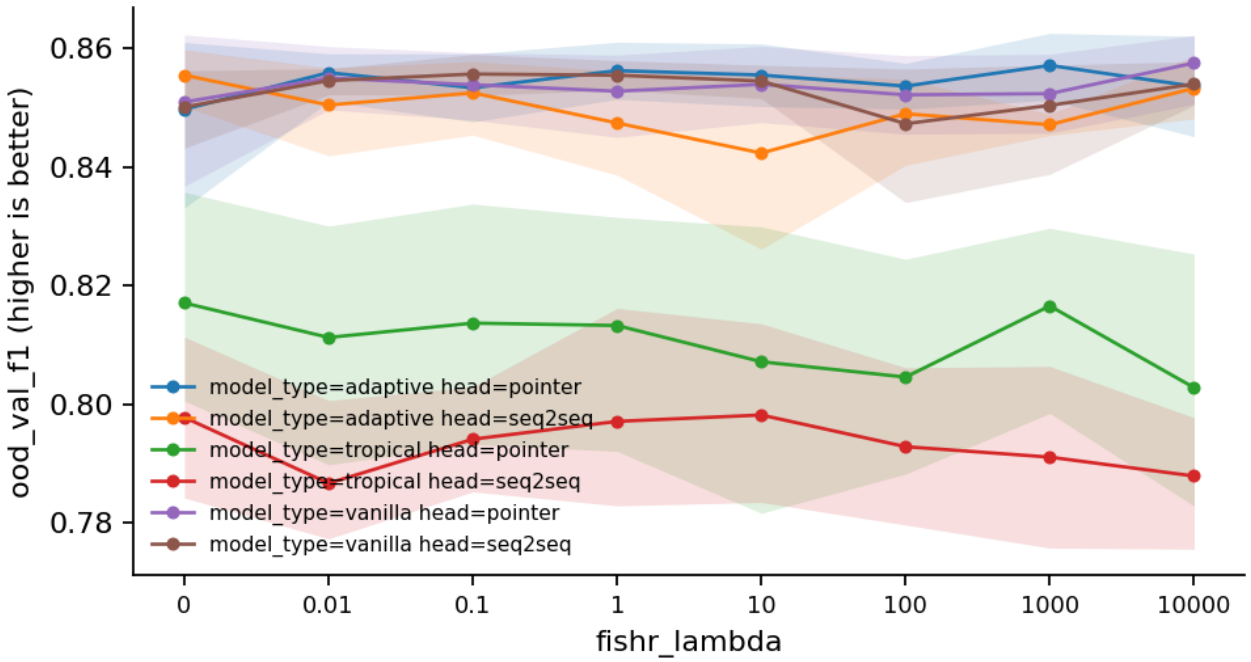


Figure 3: `ood_val_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	-0.03392	[-0.0818, 0.014]	0.2017	
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	-0.0438	[-0.0835, -0.00405]	0.2017	
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.04023	[-0.0805, -6.05×10^{-6}]	0.2017	
model_type: tropical - vanilla head=pointer fishr_lambda=1	-0.03954	[-0.0783, -8.07×10^{-4}]	0.2017	
model_type: tropical - vanilla head=pointer fishr_lambda=10	-0.04683	[-0.0918, -0.00183]	0.2017	
model_type: tropical - vanilla head=pointer fishr_lambda=100	-0.04764	[-0.0856, -0.0097]	0.2017	
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-0.03583	[-0.067, -0.00471]	0.2017	
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-0.05479	[-0.0992, -0.0103]	0.2017	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.05238	[-0.0749, -0.0298]	0.0356	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.06796	[-0.0871, -0.0488]	0.0083	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.06164	[-0.0726, -0.0507]	0.0015	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.05842	[-0.0871, -0.0297]	0.0482	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.05637	[-0.0809, -0.0318]	0.0356	*

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.05451	[-0.0859, -0.0231]	0.0767	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.05937	[-0.0842, -0.0346]	0.0346	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.06616	[-0.0835, -0.0488]	0.0068	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.001221	[-0.0338, 0.0314]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	9.024×10^{-4}	[-0.0122, 0.014]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-5.121×10^{-4}	[-0.0134, 0.0124]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.003477	[-0.00864, 0.0156]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.001563	[-0.00649, 0.00962]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.00147	[-0.00949, 0.0124]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.004766	[-0.00924, 0.0188]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.003937	[-0.0184, 0.0106]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.005388	[-0.0124, 0.0231]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.004119	[-0.0173, 0.00909]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.00319	[-0.0159, 0.00954]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.00806	[-0.0239, 0.00775]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.01216	[-0.0378, 0.0135]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.001684	[-0.0246, 0.028]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.003257	[-0.0189, 0.0124]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-7.115×10^{-4}	[-0.00886, 0.00744]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.005756	[-0.0311, 0.0196]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.005475	[-0.0105, 0.0214]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	8.667×10^{-4}	[-0.0142, 0.0159]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.008838	[-0.00708, 0.0248]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.01317	[-0.0153, 0.0416]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.004664	[-0.0111, 0.0205]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.01001	$[-6.13 \times 10^{-4}$, 0.0206]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	3.147×10^{-4}	[-0.011, 0.0116]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.01931	[-0.026, 0.0646]	1.0000	

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.02462	[-0.0142, 0.0635]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.01959	[-0.0167, 0.0558]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.01618	[-0.0396, 0.072]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.008993	[-0.018, 0.036]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.01174	[-0.0276, 0.0511]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.02552	[-0.00898, 0.06]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.01491	[-0.0366, 0.0665]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	8.531×10^{-4}	[-0.0273, 0.029]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	4.531×10^{-4}	[-0.00602, 0.00693]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.001811	[-0.00793, 0.0043]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.0027	[-0.00981, 0.00441]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-5.501×10^{-4}	[-0.00882, 0.00772]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.004878	[-0.0187, 0.0285]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.001987	[-0.0185, 0.0225]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.00354	[-0.00443, 0.0115]	1.0000

A.3 `ood_val_exact`

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.321	[0.293, 0.345]	0.0322	5	0
adaptive	pointer	0.01	0.3193	[0.303, 0.334]	0.0203	5	0
adaptive	pointer	0.1	0.3145	[0.296, 0.333]	0.0251	5	0
adaptive	pointer	1	0.3203	[0.3, 0.34]	0.0257	5	0
adaptive	pointer	10	0.3194	[0.299, 0.34]	0.0268	5	0
adaptive	pointer	100	0.3151	[0.299, 0.331]	0.0217	5	0
adaptive	pointer	1000	0.3253	[0.302, 0.349]	0.0301	5	0
adaptive	pointer	1e+04	0.3132	[0.288, 0.339]	0.0335	5	0
adaptive	seq2seq	0	0.324	[0.316, 0.334]	0.0122	5	0
adaptive	seq2seq	0.01	0.3172	[0.294, 0.334]	0.0266	5	0
adaptive	seq2seq	0.1	0.3176	[0.296, 0.33]	0.0243	5	0
adaptive	seq2seq	1	0.3135	[0.282, 0.334]	0.0344	5	0
adaptive	seq2seq	10	0.2947	[0.248, 0.325]	0.052	5	0
adaptive	seq2seq	100	0.3111	[0.292, 0.33]	0.025	5	0
adaptive	seq2seq	1000	0.3177	[0.313, 0.322]	0.00557	5	0
adaptive	seq2seq	1e+04	0.2995	[0.279, 0.32]	0.0269	5	0
tropical	pointer	0	0.2187	[0.178, 0.264]	0.0568	5	0
tropical	pointer	0.01	0.1994	[0.15, 0.249]	0.065	5	0

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	pointer	0.1	0.2062	[0.156, 0.251]	0.0635	5	0
tropical	pointer	1	0.2045	[0.152, 0.255]	0.0666	5	0
tropical	pointer	10	0.1884	[0.132, 0.245]	0.0735	5	0
tropical	pointer	100	0.1876	[0.156, 0.233]	0.0519	5	0
tropical	pointer	1000	0.2081	[0.165, 0.244]	0.0514	5	0
tropical	pointer	1e+04	0.1918	[0.133, 0.252]	0.0779	5	0
tropical	seq2seq	0	0.1391	[0.114, 0.164]	0.0316	5	0
tropical	seq2seq	0.01	0.1266	[0.104, 0.153]	0.0313	5	0
tropical	seq2seq	0.1	0.1427	[0.124, 0.16]	0.0236	5	0
tropical	seq2seq	1	0.1473	[0.113, 0.194]	0.052	5	0
tropical	seq2seq	10	0.1501	[0.119, 0.183]	0.0412	5	0
tropical	seq2seq	100	0.1326	[0.1, 0.173]	0.0454	5	0
tropical	seq2seq	1000	0.1419	[0.0953, 0.19]	0.0613	5	0
tropical	seq2seq	1e+04	0.1195	[0.0999, 0.139]	0.0256	5	0
vanilla	pointer	0	0.3166	[0.284, 0.341]	0.0384	5	0
vanilla	pointer	0.01	0.3197	[0.308, 0.33]	0.0147	5	0
vanilla	pointer	0.1	0.3149	[0.304, 0.324]	0.013	5	0
vanilla	pointer	1	0.314	[0.302, 0.324]	0.0138	5	0
vanilla	pointer	10	0.317	[0.3, 0.328]	0.0194	5	0
vanilla	pointer	100	0.3106	[0.293, 0.323]	0.0195	5	0
vanilla	pointer	1000	0.3131	[0.297, 0.327]	0.0194	5	0
vanilla	pointer	1e+04	0.3251	[0.31, 0.339]	0.0191	5	0
vanilla	seq2seq	0	0.303	[0.283, 0.323]	0.0264	5	0
vanilla	seq2seq	0.01	0.3199	[0.308, 0.333]	0.0158	5	0
vanilla	seq2seq	0.1	0.3244	[0.31, 0.336]	0.016	5	0
vanilla	seq2seq	1	0.3217	[0.311, 0.332]	0.0137	5	0
vanilla	seq2seq	10	0.3173	[0.307, 0.328]	0.0138	5	0
vanilla	seq2seq	100	0.3058	[0.289, 0.322]	0.0225	5	0
vanilla	seq2seq	1000	0.3133	[0.294, 0.331]	0.0239	5	0
vanilla	seq2seq	1e+04	0.2995	[0.287, 0.314]	0.0182	5	0

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-0.00168	[-0.0493, 0.046]	1.0000
adaptive	pointer	0.1	-0.00642	[-0.0541, 0.0412]	0.9992
adaptive	pointer	1	-6.200×10^{-4}	[-0.0483, 0.047]	1.0000
adaptive	pointer	10	-0.00158	[-0.0492, 0.0461]	1.0000
adaptive	pointer	100	-0.00584	[-0.0535, 0.0418]	0.9996
adaptive	pointer	1000	0.00438	[-0.0433, 0.052]	0.9999
adaptive	pointer	1e+04	-0.0078	[-0.0555, 0.0399]	0.9975
adaptive	seq2seq	0.01	-0.00688	[-0.0575, 0.0437]	0.9992
adaptive	seq2seq	0.1	-0.00646	[-0.0571, 0.0441]	0.9995
adaptive	seq2seq	1	-0.0105	[-0.0611, 0.0401]	0.9898
adaptive	seq2seq	10	-0.0293	[-0.0799, 0.0213]	0.4523
adaptive	seq2seq	100	-0.01292	[-0.0635, 0.0377]	0.9692
adaptive	seq2seq	1000	-0.0063	[-0.0569, 0.0443]	0.9995
adaptive	seq2seq	1e+04	-0.02454	[-0.0751, 0.0261]	0.6308
tropical	pointer	0.01	-0.01924	[-0.131, 0.0925]	0.9966
tropical	pointer	0.1	-0.0125	[-0.124, 0.0992]	0.9998
tropical	pointer	1	-0.01414	[-0.126, 0.0976]	0.9995
tropical	pointer	10	-0.03032	[-0.142, 0.0814]	0.9584
tropical	pointer	100	-0.03106	[-0.143, 0.0807]	0.9532

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
tropical	pointer	1000	-0.0106	[-0.122, 0.101]	0.9999
tropical	pointer	1e+04	-0.0269	[-0.139, 0.0848]	0.9772
tropical	seq2seq	0.01	-0.0125	[-0.084, 0.059]	0.9962
tropical	seq2seq	0.1	0.00362	[-0.0679, 0.0751]	1.0000
tropical	seq2seq	1	0.00822	[-0.0633, 0.0797]	0.9997
tropical	seq2seq	10	0.01096	[-0.0606, 0.0825]	0.9983
tropical	seq2seq	100	-0.00652	[-0.078, 0.065]	0.9999
tropical	seq2seq	1000	0.00282	[-0.0687, 0.0743]	1.0000
tropical	seq2seq	1e+04	-0.0196	[-0.0911, 0.0519]	0.9564
vanilla	pointer	0.01	0.00306	[-0.0337, 0.0398]	1.0000
vanilla	pointer	0.1	-0.00172	[-0.0385, 0.035]	1.0000
vanilla	pointer	1	-0.00258	[-0.0393, 0.0342]	1.0000
vanilla	pointer	10	3.800×10^{-4}	[-0.0364, 0.0371]	1.0000
vanilla	pointer	100	-0.006	[-0.0428, 0.0308]	0.9975
vanilla	pointer	1000	-0.0035	[-0.0403, 0.0333]	0.9999
vanilla	pointer	1e+04	0.00854	[-0.0282, 0.0453]	0.9811
vanilla	seq2seq	0.01	0.01698	[-0.0168, 0.0507]	0.5950
vanilla	seq2seq	0.1	0.0214	[-0.0123, 0.0551]	0.3601
vanilla	seq2seq	1	0.01878	[-0.015, 0.0525]	0.4932
vanilla	seq2seq	10	0.01438	[-0.0194, 0.0481]	0.7445
vanilla	seq2seq	100	0.00286	[-0.0309, 0.0366]	1.0000
vanilla	seq2seq	1000	0.01038	[-0.0234, 0.0441]	0.9255
vanilla	seq2seq	1e+04	-0.00344	[-0.0372, 0.0303]	0.9999

Table 15: `ood_val_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-7.071×10^{-5}	[-0.00231, 0.00226]	0.9732	-0.0019	1.0000
adaptive	seq2seq	-0.001967	[-0.00457, 7.92×10^{-4}]	0.4327	-0.17	0.2038
tropical	pointer	-0.001289	[-0.00725, 0.00304]	0.8019	-0.0362	0.7952
tropical	seq2seq	-0.001344	[-0.00452, 0.00187]	0.6947	-0.099	0.4671
vanilla	pointer	3.379×10^{-4}	[-0.00102, 0.0017]	0.8099	0.0648	0.6453
vanilla	seq2seq	-0.003544	[-0.00692, -0.00118]	0.0209	-0.29	0.0340

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	-0.09792	[-0.212, 0.0163]	0.1478
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	-0.1202	[-0.215, -0.0252]	0.1478
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.1087	[-0.2, -0.0175]	0.1478
model_type: tropical - vanilla head=pointer fishr_lambda=1	-0.1095	[-0.206, -0.0131]	0.1478
model_type: tropical - vanilla head=pointer fishr_lambda=10	-0.1286	[-0.234, -0.0233]	0.1478
model_type: tropical - vanilla head=pointer fishr_lambda=100	-0.123	[-0.211, -0.0347]	0.1444

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-0.105	[-0.184, -0.0262]	0.1460	
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-0.1334	[-0.251, -0.0158]	0.1478	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.1639	[-0.196, -0.132]	0.0021	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.1933	[-0.241, -0.146]	0.0045	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.1816	[-0.213, -0.151]	0.0013	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.1744	[-0.235, -0.114]	0.0142	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.1673	[-0.223, -0.112]	0.0134	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.1732	[-0.242, -0.105]	0.0193	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.1714	[-0.236, -0.107]	0.0183	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.18	[-0.214, -0.146]	0.0018	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.00436	[-0.0525, 0.0612]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-3.800×10^{-4}	[-0.0346, 0.0339]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-3.400×10^{-4}	[-0.0263, 0.0257]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.00632	[-0.025, 0.0377]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.0024	[-0.0169, 0.0217]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.00452	[-0.0191, 0.0281]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.01224	[-0.0301, 0.0546]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.01198	[-0.0409, 0.0169]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.02108	[-0.0229, 0.0651]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.00278	[-0.0335, 0.0279]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.00678	[-0.0472, 0.0336]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.0082	[-0.0549, 0.0385]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.0226	[-0.0782, 0.033]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.0053	[-0.04, 0.0506]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.0044	[-0.0316, 0.0404]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-2.000×10^{-5}	[-0.0235, 0.0234]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.00308	[-0.0558, 0.0496]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.00212	[-0.0515, 0.0557]	1.0000	

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.00304	[-0.0549, 0.0488]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.0068	[-0.0547, 0.0683]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.02464	[-0.0572, 0.106]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.004	[-0.0452, 0.0532]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.0076	[-0.0358, 0.051]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.01366	[-0.0206, 0.0479]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.07958	[-0.0227, 0.182]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.07284	[-0.0237, 0.169]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.06346	[-0.0172, 0.144]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.05722	[-0.0744, 0.189]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.0383	[-0.0229, 0.0995]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.05504	[-0.0331, 0.143]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.06616	[-0.0359, 0.168]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.07228	[-0.0463, 0.191]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.01364	[-0.0543, 0.0816]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-2.800×10^{-4}	[-0.00852, 0.00796]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.00948	[-0.0147, -0.00429]	0.1715
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.00772	[-0.0121, -0.00335]	0.1849
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-3.600×10^{-4}	[-0.0236, 0.0229]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.00478	[-0.0256, 0.0351]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-2.400×10^{-4}	[-0.0359, 0.0355]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.02562	[-0.00431, 0.0555]	1.0000

A.4 `ood_noise_f1`

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.6519	[0.634, 0.67]	0.0241	5	0
adaptive	pointer	0.01	0.6557	[0.648, 0.664]	0.0102	5	0
adaptive	pointer	0.1	0.6514	[0.645, 0.663]	0.0129	5	0
adaptive	pointer	1	0.6492	[0.643, 0.659]	0.0109	5	0

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	10	0.6539	[0.641, 0.669]	0.0175	5	0
adaptive	pointer	100	0.6545	[0.643, 0.669]	0.0156	5	0
adaptive	pointer	1000	0.6523	[0.634, 0.678]	0.0288	5	0
adaptive	pointer	1e+04	0.6656	[0.644, 0.68]	0.0238	5	0
adaptive	seq2seq	0	0.583	[0.571, 0.597]	0.0171	5	0
adaptive	seq2seq	0.01	0.5916	[0.57, 0.612]	0.0271	5	0
adaptive	seq2seq	0.1	0.5874	[0.57, 0.605]	0.0234	5	0
adaptive	seq2seq	1	0.5699	[0.542, 0.592]	0.0336	5	0
adaptive	seq2seq	10	0.5972	[0.578, 0.609]	0.022	5	0
adaptive	seq2seq	100	0.6266	[0.591, 0.66]	0.0448	5	0
adaptive	seq2seq	1000	0.5774	[0.525, 0.62]	0.0618	5	0
adaptive	seq2seq	1e+04	0.3681	[0.281, 0.455]	0.11	5	0
tropical	pointer	0	0.6814	[0.662, 0.698]	0.0225	5	0
tropical	pointer	0.01	0.6702	[0.637, 0.704]	0.0436	5	0
tropical	pointer	0.1	0.6847	[0.661, 0.708]	0.0309	5	0
tropical	pointer	1	0.6867	[0.665, 0.709]	0.0283	5	0
tropical	pointer	10	0.6756	[0.649, 0.702]	0.0352	5	0
tropical	pointer	100	0.6587	[0.613, 0.704]	0.0587	5	0
tropical	pointer	1000	0.6757	[0.659, 0.694]	0.0227	5	0
tropical	pointer	1e+04	0.6744	[0.651, 0.693]	0.0279	5	0
tropical	seq2seq	0	0.4885	[0.361, 0.639]	0.172	5	0
tropical	seq2seq	0.01	0.5831	[0.478, 0.699]	0.138	5	0
tropical	seq2seq	0.1	0.5602	[0.435, 0.685]	0.16	5	0
tropical	seq2seq	1	0.5357	[0.391, 0.678]	0.198	5	0
tropical	seq2seq	10	0.5538	[0.432, 0.676]	0.159	5	0
tropical	seq2seq	100	0.5943	[0.434, 0.708]	0.191	5	0
tropical	seq2seq	1000	0.5792	[0.436, 0.7]	0.168	5	0
tropical	seq2seq	1e+04	0.5096	[0.418, 0.625]	0.134	5	0
vanilla	pointer	0	0.6196	[0.572, 0.665]	0.0597	5	0
vanilla	pointer	0.01	0.6284	[0.583, 0.673]	0.0628	5	0
vanilla	pointer	0.1	0.6222	[0.57, 0.67]	0.0634	5	0
vanilla	pointer	1	0.6382	[0.605, 0.672]	0.0428	5	0
vanilla	pointer	10	0.6261	[0.576, 0.669]	0.0608	5	0
vanilla	pointer	100	0.6244	[0.567, 0.673]	0.0671	5	0
vanilla	pointer	1000	0.6192	[0.577, 0.661]	0.0544	5	0
vanilla	pointer	1e+04	0.632	[0.587, 0.668]	0.0501	5	0
vanilla	seq2seq	0	0.5557	[0.49, 0.601]	0.0748	5	0
vanilla	seq2seq	0.01	0.5761	[0.562, 0.587]	0.0158	5	0
vanilla	seq2seq	0.1	0.5828	[0.561, 0.606]	0.029	5	0
vanilla	seq2seq	1	0.5834	[0.561, 0.612]	0.0321	5	0
vanilla	seq2seq	10	0.5787	[0.55, 0.605]	0.0346	5	0
vanilla	seq2seq	100	0.5892	[0.567, 0.619]	0.0338	5	0
vanilla	seq2seq	1000	0.5731	[0.541, 0.617]	0.0518	5	0
vanilla	seq2seq	1e+04	0.3824	[0.313, 0.453]	0.0936	5	0

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0.003766	[-0.0296, 0.0371]	0.9998
adaptive	pointer	0.1	-4.889×10^{-4}	[-0.0338, 0.0329]	1.0000
adaptive	pointer	1	-0.002664	[-0.036, 0.0307]	1.0000
adaptive	pointer	10	0.002013	[-0.0313, 0.0354]	1.0000
adaptive	pointer	100	0.002612	[-0.0307, 0.036]	1.0000

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	1000	3.760×10^{-4}	[-0.033, 0.0337]	1.0000	
adaptive	pointer	1e+04	0.01367	[-0.0197, 0.047]	0.7745	
adaptive	seq2seq	0.01	0.008691	[-0.0808, 0.0982]	0.9999	
adaptive	seq2seq	0.1	0.00442	[-0.0851, 0.0939]	1.0000	
adaptive	seq2seq	1	-0.01303	[-0.103, 0.0765]	0.9988	
adaptive	seq2seq	10	0.0143	[-0.0752, 0.104]	0.9978	
adaptive	seq2seq	100	0.04361	[-0.0459, 0.133]	0.6262	
adaptive	seq2seq	1000	-0.00556	[-0.0951, 0.0839]	1.0000	
adaptive	seq2seq	1e+04	-0.2148	[-0.304, -0.125]	$< 10^{-4}$	*
tropical	pointer	0.01	-0.01119	[-0.0734, 0.051]	0.9955	
tropical	pointer	0.1	0.003271	[-0.0589, 0.0655]	1.0000	
tropical	pointer	1	0.00533	[-0.0569, 0.0675]	1.0000	
tropical	pointer	10	-0.00577	[-0.068, 0.0564]	0.9999	
tropical	pointer	100	-0.02271	[-0.0849, 0.0395]	0.8496	
tropical	pointer	1000	-0.005648	[-0.0678, 0.0565]	0.9999	
tropical	pointer	1e+04	-0.007029	[-0.0692, 0.0552]	0.9998	
tropical	seq2seq	0.01	0.09466	[-0.196, 0.385]	0.9043	
tropical	seq2seq	0.1	0.07179	[-0.219, 0.362]	0.9739	
tropical	seq2seq	1	0.04726	[-0.243, 0.338]	0.9976	
tropical	seq2seq	10	0.06534	[-0.225, 0.356]	0.9841	
tropical	seq2seq	100	0.1058	[-0.185, 0.396]	0.8510	
tropical	seq2seq	1000	0.09076	[-0.2, 0.381]	0.9201	
tropical	seq2seq	1e+04	0.02117	[-0.269, 0.312]	1.0000	
vanilla	pointer	0.01	0.008854	[-0.0927, 0.11]	1.0000	
vanilla	pointer	0.1	0.002653	[-0.0989, 0.104]	1.0000	
vanilla	pointer	1	0.01859	[-0.083, 0.12]	0.9950	
vanilla	pointer	10	0.006473	[-0.0951, 0.108]	1.0000	
vanilla	pointer	100	0.004796	[-0.0967, 0.106]	1.0000	
vanilla	pointer	1000	-3.701×10^{-4}	[-0.102, 0.101]	1.0000	
vanilla	pointer	1e+04	0.01244	[-0.0891, 0.114]	0.9996	
vanilla	seq2seq	0.01	0.02043	[-0.0701, 0.111]	0.9839	
vanilla	seq2seq	0.1	0.02711	[-0.0634, 0.118]	0.9340	
vanilla	seq2seq	1	0.02765	[-0.0629, 0.118]	0.9278	
vanilla	seq2seq	10	0.02298	[-0.0676, 0.114]	0.9701	
vanilla	seq2seq	100	0.03344	[-0.0571, 0.124]	0.8432	
vanilla	seq2seq	1000	0.01734	[-0.0732, 0.108]	0.9936	
vanilla	seq2seq	1e+04	-0.1733	[-0.264, -0.0828]	$< 10^{-4}$	*

Table 19: `ood_noise_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.001311	[-0.00136, 0.00337]	0.3776	0.059	0.6703
adaptive	seq2seq	-0.02264	[-0.0337, -0.0116]	0.0032	-0.192	0.1494
tropical	pointer	-0.001192	[-0.00392, 0.00241]	0.6913	-0.059	0.6753
tropical	seq2seq	-0.004428	[-0.00748, -0.00138]	0.7370	-0.0476	0.7261
vanilla	pointer	-3.243×10^{-4}	[-0.00321, 0.00317]	0.9435	-0.0133	0.9346
vanilla	seq2seq	-0.02125	[-0.0287, -0.0138]	0.0013	-0.387	0.0026

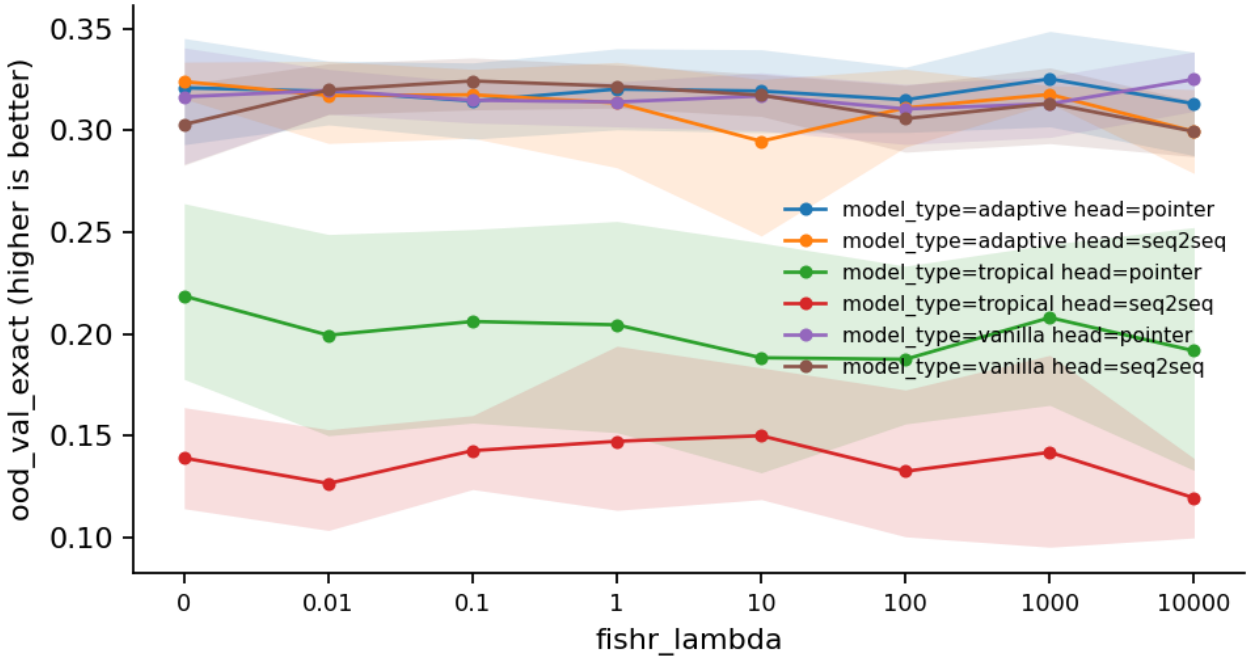


Figure 4: `ood_val_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

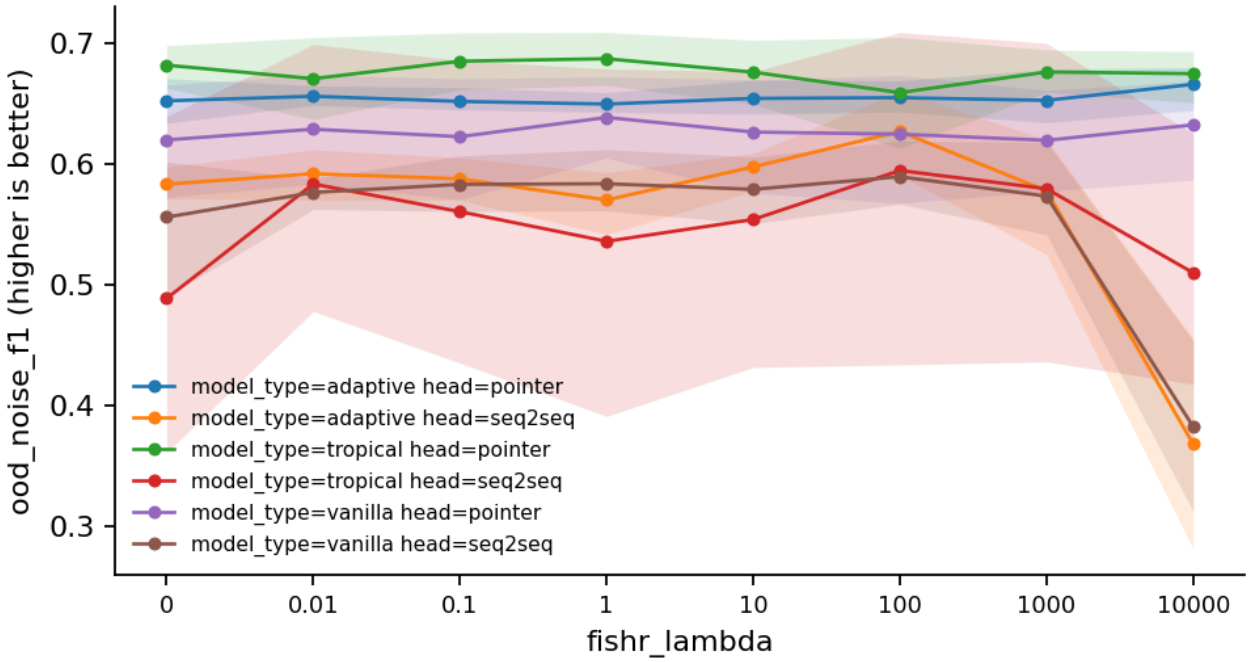


Figure 5: `ood_noise_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.0618	[0.00317, 0.12]	0.6874
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.04176	[-0.0591, 0.143]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.06242	[-0.0276, 0.152]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.04854	[-0.0252, 0.122]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.04956	[-0.0345, 0.134]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.0343	[-0.0593, 0.128]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.05653	[-0.00631, 0.119]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.04233	[-0.0418, 0.127]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.06726	[-0.275, 0.14]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.006979	[-0.173, 0.187]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.02257	[-0.229, 0.184]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.04765	[-0.307, 0.211]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.0249	[-0.235, 0.185]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.005107	[-0.24, 0.25]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.006172	[-0.175, 0.187]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.1272	[-0.104, 0.358]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.03231	[-0.0414, 0.106]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.02722	[-0.059, 0.113]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.02917	[-0.0633, 0.122]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.01106	[-0.0483, 0.0704]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.02785	[-0.0631, 0.119]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.03013	[-0.0683, 0.129]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.03306	[-0.0546, 0.121]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.03354	[-0.0101, 0.0772]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.02724	[-0.0857, 0.14]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.0155	[-0.0327, 0.0637]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.004545	[-0.0499, 0.059]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.01344	[-0.0897, 0.0628]	1.0000

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.01856	[-0.0292, 0.0664]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.03741	[-0.027, 0.102]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.00434	[-0.0685, 0.0771]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.0143	[-0.0972, 0.0686]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.06894	[0.0351, 0.103]	0.1139
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.06402	[0.0327, 0.0953]	0.1139
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.06403	[0.0325, 0.0956]	0.1139
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.0793	[0.0394, 0.119]	0.1139
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.05666	[0.0267, 0.0866]	0.1189
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.02795	[-0.0236, 0.0795]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.07488	[-0.0069, 0.157]	0.9576
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.2974	[0.144, 0.451]	0.1163
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.1929	[-0.0312, 0.417]	0.9773
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.08707	[-0.12, 0.294]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.1244	[-0.0909, 0.34]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.151	[-0.0988, 0.401]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.1218	[-0.105, 0.349]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.06442	[-0.211, 0.34]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.09652	[-0.13, 0.323]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.1647	[-0.0107, 0.34]	0.9537
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.06387	[-0.0782, 0.206]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.05229	[-0.0108, 0.115]	0.9940
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.03941	[-0.0262, 0.105]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.0548	$[-2.42 \times 10^{-4}, 0.11]$	0.8607
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.04736	[-0.00583, 0.101]	0.9631
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.03523	[-0.039, 0.109]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.04616	[-0.064, 0.156]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.2496	[0.106, 0.393]	0.1527

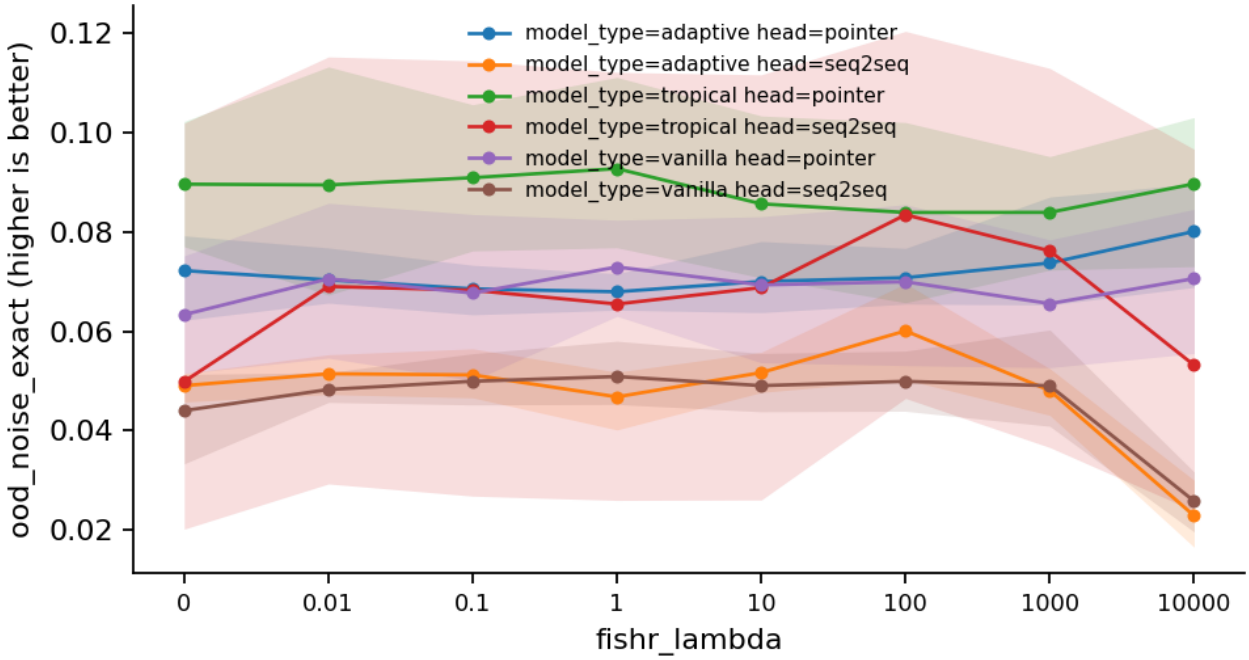


Figure 6: `ood_noise_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A.5 `ood_noise_exact`

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.07218	[0.0622, 0.0792]	0.0111	5	0
adaptive	pointer	0.01	0.07032	[0.0657, 0.0768]	0.00735	5	0
adaptive	pointer	0.1	0.06852	[0.0633, 0.0733]	0.00627	5	0
adaptive	pointer	1	0.06792	[0.0642, 0.0716]	0.00476	5	0
adaptive	pointer	10	0.07	[0.0638, 0.0781]	0.00901	5	0
adaptive	pointer	100	0.07076	[0.0655, 0.0767]	0.00722	5	0
adaptive	pointer	1000	0.07372	[0.0653, 0.087]	0.0144	5	0
adaptive	pointer	1e+04	0.08006	[0.0688, 0.0896]	0.014	5	0
adaptive	seq2seq	0	0.04904	[0.0458, 0.0515]	0.00381	5	0
adaptive	seq2seq	0.01	0.05144	[0.0473, 0.0553]	0.00525	5	0
adaptive	seq2seq	0.1	0.0512	[0.0466, 0.0565]	0.00629	5	0
adaptive	seq2seq	1	0.04678	[0.0402, 0.0518]	0.00732	5	0
adaptive	seq2seq	10	0.05166	[0.0477, 0.0558]	0.00526	5	0
adaptive	seq2seq	100	0.06006	[0.0503, 0.0696]	0.0129	5	0
adaptive	seq2seq	1000	0.04806	[0.0431, 0.0525]	0.00597	5	0
adaptive	seq2seq	1e+04	0.02288	[0.0166, 0.03]	0.0088	5	0
tropical	pointer	0	0.08958	[0.077, 0.102]	0.0159	5	0
tropical	pointer	0.01	0.08942	[0.0673, 0.113]	0.0299	5	0
tropical	pointer	0.1	0.09088	[0.0762, 0.106]	0.0188	5	0
tropical	pointer	1	0.0927	[0.0768, 0.111]	0.023	5	0
tropical	pointer	10	0.08562	[0.0708, 0.103]	0.0212	5	0
tropical	pointer	100	0.08388	[0.0658, 0.102]	0.0249	5	0
tropical	pointer	1000	0.0839	[0.0723, 0.0951]	0.0146	5	0
tropical	pointer	1e+04	0.08964	[0.073, 0.103]	0.0191	5	0
tropical	seq2seq	0	0.04986	[0.0202, 0.102]	0.0573	5	0

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	0.01	0.069	[0.0293, 0.115]	0.0544	5	0
tropical	seq2seq	0.1	0.06824	[0.0268, 0.114]	0.0559	5	0
tropical	seq2seq	1	0.06546	[0.026, 0.112]	0.056	5	0
tropical	seq2seq	10	0.06876	[0.0261, 0.112]	0.0594	5	0
tropical	seq2seq	100	0.08342	[0.0465, 0.12]	0.0506	5	0
tropical	seq2seq	1000	0.07616	[0.0366, 0.113]	0.0501	5	0
tropical	seq2seq	1e+04	0.05316	[0.0239, 0.0966]	0.0482	5	0
vanilla	pointer	0	0.06332	[0.0515, 0.0751]	0.0155	5	0
vanilla	pointer	0.01	0.07054	[0.0546, 0.0857]	0.0208	5	0
vanilla	pointer	0.1	0.06766	[0.0501, 0.0835]	0.0214	5	0
vanilla	pointer	1	0.07292	[0.063, 0.0823]	0.013	5	0
vanilla	pointer	10	0.0693	[0.0537, 0.083]	0.0187	5	0
vanilla	pointer	100	0.06994	[0.0531, 0.0855]	0.0215	5	0
vanilla	pointer	1000	0.06554	[0.0527, 0.0784]	0.017	5	0
vanilla	pointer	1e+04	0.07058	[0.0555, 0.0845]	0.0184	5	0
vanilla	seq2seq	0	0.04402	[0.0333, 0.0513]	0.0123	5	0
vanilla	seq2seq	0.01	0.04826	[0.0457, 0.0517]	0.00399	5	0
vanilla	seq2seq	0.1	0.04992	[0.0452, 0.0555]	0.00658	5	0
vanilla	seq2seq	1	0.05088	[0.0453, 0.058]	0.00862	5	0
vanilla	seq2seq	10	0.04904	[0.0438, 0.0555]	0.00778	5	0
vanilla	seq2seq	100	0.0499	[0.0439, 0.056]	0.00772	5	0
vanilla	seq2seq	1000	0.04902	[0.0409, 0.0603]	0.013	5	0
vanilla	seq2seq	1e+04	0.02582	[0.0196, 0.0317]	0.00768	5	0

Table 22: `ood_noise_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	-0.00186	[-0.0191, 0.0153]	0.9998	
adaptive	pointer	0.1	-0.00366	[-0.0209, 0.0135]	0.9883	
adaptive	pointer	1	-0.00426	[-0.0215, 0.0129]	0.9736	
adaptive	pointer	10	-0.00218	[-0.0194, 0.015]	0.9995	
adaptive	pointer	100	-0.00142	[-0.0186, 0.0158]	1.0000	
adaptive	pointer	1000	0.00154	[-0.0157, 0.0187]	0.9999	
adaptive	pointer	1e+04	0.00788	[-0.00932, 0.0251]	0.6835	
adaptive	seq2seq	0.01	0.0024	[-0.0106, 0.0154]	0.9947	
adaptive	seq2seq	0.1	0.00216	[-0.0108, 0.0151]	0.9972	
adaptive	seq2seq	1	-0.00226	[-0.0152, 0.0107]	0.9963	
adaptive	seq2seq	10	0.00262	[-0.0104, 0.0156]	0.9913	
adaptive	seq2seq	100	0.01102	[-0.00195, 0.024]	0.1227	
adaptive	seq2seq	1000	-9.800×10^{-4}	[-0.014, 0.012]	1.0000	
adaptive	seq2seq	1e+04	-0.02616	[-0.0391, -0.0132]	$< 10^{-4}$	*
tropical	pointer	0.01	-1.600×10^{-4}	[-0.0376, 0.0373]	1.0000	
tropical	pointer	0.1	0.0013	[-0.0361, 0.0387]	1.0000	
tropical	pointer	1	0.00312	[-0.0343, 0.0406]	1.0000	
tropical	pointer	10	-0.00396	[-0.0414, 0.0335]	0.9998	
tropical	pointer	100	-0.0057	[-0.0431, 0.0317]	0.9984	
tropical	pointer	1000	-0.00568	[-0.0431, 0.0318]	0.9984	
tropical	pointer	1e+04	6.000×10^{-5}	[-0.0374, 0.0375]	1.0000	
tropical	seq2seq	0.01	0.01914	[-0.0754, 0.114]	0.9911	
tropical	seq2seq	0.1	0.01838	[-0.0762, 0.113]	0.9930	
tropical	seq2seq	1	0.0156	[-0.0789, 0.11]	0.9973	
tropical	seq2seq	10	0.0189	[-0.0756, 0.113]	0.9918	
tropical	seq2seq	100	0.03356	[-0.061, 0.128]	0.8650	

Table 22: `ood_noise_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p	
tropical	seq2seq	1000	0.0263	[-0.0682, 0.121]	0.9531	
tropical	seq2seq	1e+04	0.0033	[-0.0912, 0.0978]	1.0000	
vanilla	pointer	0.01	0.00722	[-0.0251, 0.0396]	0.9847	
vanilla	pointer	0.1	0.00434	[-0.028, 0.0367]	0.9993	
vanilla	pointer	1	0.0096	[-0.0227, 0.0419]	0.9364	
vanilla	pointer	10	0.00598	[-0.0264, 0.0383]	0.9947	
vanilla	pointer	100	0.00662	[-0.0257, 0.039]	0.9906	
vanilla	pointer	1000	0.00222	[-0.0301, 0.0346]	1.0000	
vanilla	pointer	1e+04	0.00726	[-0.0251, 0.0396]	0.9843	
vanilla	seq2seq	0.01	0.00424	[-0.0113, 0.0198]	0.9571	
vanilla	seq2seq	0.1	0.0059	[-0.00963, 0.0214]	0.8263	
vanilla	seq2seq	1	0.00686	[-0.00867, 0.0224]	0.7150	
vanilla	seq2seq	10	0.00502	[-0.0105, 0.0206]	0.9077	
vanilla	seq2seq	100	0.00588	[-0.00965, 0.0214]	0.8283	
vanilla	seq2seq	1000	0.005	[-0.0105, 0.0205]	0.9092	
vanilla	seq2seq	1e+04	-0.0182	[-0.0337, -0.00267]	0.0162	*

Table 23: `ood_noise_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.001516	$[1.00 \times 10^{-4}$, 0.00293]	0.0604	0.173	0.1942
adaptive	seq2seq	-0.00281	[-0.00377, -0.00174]	0.0085	-0.272	0.0374
tropical	pointer	-7.900×10^{-4}	[-0.00242, 9.74×10^{-4}]	0.6527	-0.0381	0.7984
tropical	seq2seq	-4.900×10^{-4}	[-0.00209, 0.00111]	0.9088	-0.0133	0.9294
vanilla	pointer	-2.536×10^{-4}	[-0.00145, 9.45×10^{-4}]	0.8653	0.0019	0.9980
vanilla	seq2seq	-0.002504	[-0.00335, -0.00163]	0.0068	-0.333	0.0120

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.02626	[-0.00289, 0.0554]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.01888	[-0.0311, 0.0689]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.02322	[-0.0153, 0.0617]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.01978	[-0.022, 0.0616]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.01632	[-0.0231, 0.0557]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.01394	[-0.0268, 0.0547]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.01836	[-0.0159, 0.0526]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.01906	[-0.021, 0.0591]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.00584	[-0.0603, 0.072]	1.0000

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.02074	[-0.0486, 0.0901]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.01832	[-0.0512, 0.0878]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.01458	[-0.0607, 0.0898]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.01972	[-0.0551, 0.0945]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.03352	[-0.0329, 0.1]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.02714	[-0.0304, 0.0847]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.02734	[-0.034, 0.0887]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.00886	[-0.02, 0.0377]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-2.200×10^{-4}	[-0.0258, 0.0254]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	8.600×10^{-4}	[-0.0289, 0.0306]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.005	[-0.0234, 0.0134]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	7.000×10^{-4}	[-0.0242, 0.0256]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	8.200×10^{-4}	[-0.0272, 0.0289]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.00818	[-0.0268, 0.0432]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.00948	[-0.00985, 0.0288]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.00502	[-0.0129, 0.023]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.00318	[-0.00333, 0.00969]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.00128	[-0.0113, 0.0138]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.0041	[-0.0226, 0.0144]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.00262	[-0.00914, 0.0144]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.01016	[-0.0109, 0.0312]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-9.600×10^{-4}	[-0.0166, 0.0147]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.00294	[-0.01, 0.00412]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.02314	[0.00929, 0.037]	0.2240
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.01888	[0.0065, 0.0313]	0.2546
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.01732	[0.00459, 0.03]	0.3309
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.02114	[0.00766, 0.0346]	0.2546
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.01834	[0.0047, 0.032]	0.3309

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.0107	[-0.00852, 0.0299]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.02566	[0.00968, 0.0416]	0.2460
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.05718	[0.0294, 0.085]	0.1117
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.03972	[-0.0317, 0.111]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.02042	[-0.0662, 0.107]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.02264	[-0.0532, 0.0985]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.02724	[-0.0319, 0.0864]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.01686	[-0.0674, 0.101]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	4.600×10^{-4}	[-0.0692, 0.0701]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.00774	[-0.0512, 0.0666]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.03648	[-0.0302, 0.103]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.0193	[-0.00963, 0.0482]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.02228	[-0.00159, 0.0461]	0.7873
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.01774	[-0.00557, 0.0411]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.02204	[0.0079, 0.0362]	0.2546
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.02026	[0.00314, 0.0374]	0.4547
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.02004	[8.43×10^{-4}, 0.0392]	0.6186
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.01652	[-0.0127, 0.0457]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.04476	[0.0153, 0.0742]	0.2546

A.6 `id_f1`

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.8813	[0.88, 0.883]	0.0018	5	0
adaptive	pointer	0.01	0.8827	[0.881, 0.884]	0.00183	5	0
adaptive	pointer	0.1	0.8828	[0.882, 0.884]	0.0018	5	0
adaptive	pointer	1	0.8826	[0.881, 0.884]	0.00185	5	0
adaptive	pointer	10	0.8827	[0.881, 0.884]	0.00214	5	0
adaptive	pointer	100	0.8824	[0.881, 0.884]	0.00202	5	0
adaptive	pointer	1000	0.8827	[0.882, 0.884]	0.00145	5	0
adaptive	pointer	1e+04	0.8827	[0.881, 0.884]	0.00179	5	0
adaptive	seq2seq	0	0.8728	[0.871, 0.874]	0.00198	5	0
adaptive	seq2seq	0.01	0.873	[0.872, 0.875]	0.00206	5	0

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	seq2seq	0.1	0.8732	[0.872, 0.875]	0.00223	5	0
adaptive	seq2seq	1	0.8728	[0.871, 0.875]	0.00239	5	0
adaptive	seq2seq	10	0.8728	[0.871, 0.874]	0.0021	5	0
adaptive	seq2seq	100	0.8726	[0.871, 0.874]	0.00224	5	0
adaptive	seq2seq	1000	0.8711	[0.87, 0.872]	0.0012	5	0
adaptive	seq2seq	1e+04	0.8646	[0.863, 0.866]	0.00229	5	0
tropical	pointer	0	0.8798	[0.877, 0.883]	0.00433	5	0
tropical	pointer	0.01	0.882	[0.879, 0.885]	0.00404	5	0
tropical	pointer	0.1	0.8817	[0.879, 0.885]	0.00414	5	0
tropical	pointer	1	0.8817	[0.879, 0.885]	0.00417	5	0
tropical	pointer	10	0.8813	[0.878, 0.885]	0.00417	5	0
tropical	pointer	100	0.8821	[0.879, 0.886]	0.00449	5	0
tropical	pointer	1000	0.8818	[0.879, 0.886]	0.00464	5	0
tropical	pointer	1e+04	0.8825	[0.879, 0.887]	0.00503	5	0
tropical	seq2seq	0	0.8806	[0.879, 0.883]	0.00255	5	0
tropical	seq2seq	0.01	0.8819	[0.88, 0.884]	0.00262	5	0
tropical	seq2seq	0.1	0.8816	[0.88, 0.884]	0.00278	5	0
tropical	seq2seq	1	0.8815	[0.88, 0.883]	0.0022	5	0
tropical	seq2seq	10	0.8814	[0.879, 0.883]	0.0029	5	0
tropical	seq2seq	100	0.8806	[0.879, 0.882]	0.00226	5	0
tropical	seq2seq	1000	0.8793	[0.877, 0.882]	0.00316	5	0
tropical	seq2seq	1e+04	0.8712	[0.869, 0.873]	0.00219	5	0
vanilla	pointer	0	0.8771	[0.877, 0.877]	3.95×10^{-4}	5	0
vanilla	pointer	0.01	0.8777	[0.877, 0.878]	6.14×10^{-4}	5	0
vanilla	pointer	0.1	0.8775	[0.877, 0.878]	8.84×10^{-4}	5	0
vanilla	pointer	1	0.8774	[0.877, 0.878]	7.44×10^{-4}	5	0
vanilla	pointer	10	0.8778	[0.877, 0.878]	9.02×10^{-4}	5	0
vanilla	pointer	100	0.8779	[0.877, 0.879]	0.00108	5	0
vanilla	pointer	1000	0.8781	[0.877, 0.879]	0.00129	5	0
vanilla	pointer	1e+04	0.8778	[0.877, 0.878]	9.84×10^{-4}	5	0
vanilla	seq2seq	0	0.8719	[0.871, 0.873]	0.00135	5	0
vanilla	seq2seq	0.01	0.873	[0.872, 0.874]	0.00132	5	0
vanilla	seq2seq	0.1	0.873	[0.872, 0.874]	0.00142	5	0
vanilla	seq2seq	1	0.8731	[0.872, 0.874]	0.00116	5	0
vanilla	seq2seq	10	0.873	[0.872, 0.874]	0.00173	5	0
vanilla	seq2seq	100	0.8732	[0.872, 0.875]	0.00193	5	0
vanilla	seq2seq	1000	0.8709	[0.87, 0.872]	6.78×10^{-4}	5	0
vanilla	seq2seq	1e+04	0.8634	[0.861, 0.866]	0.00323	5	0

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0.001371	[-0.00185, 0.00459]	0.7458
adaptive	pointer	0.1	0.001491	[-0.00173, 0.00471]	0.6742
adaptive	pointer	1	0.00131	[-0.00191, 0.00453]	0.7804
adaptive	pointer	10	0.001393	[-0.00183, 0.00462]	0.7328
adaptive	pointer	100	0.001048	[-0.00217, 0.00427]	0.9051
adaptive	pointer	1000	0.001394	[-0.00183, 0.00462]	0.7322
adaptive	pointer	1e+04	0.001352	[-0.00187, 0.00457]	0.7569
adaptive	seq2seq	0.01	2.477×10^{-4}	[-0.00341, 0.0039]	1.0000
adaptive	seq2seq	0.1	4.762×10^{-4}	[-0.00318, 0.00413]	0.9994
adaptive	seq2seq	1	1.276×10^{-5}	[-0.00364, 0.00367]	1.0000
adaptive	seq2seq	10	2.783×10^{-5}	[-0.00363, 0.00368]	1.0000

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	seq2seq	100	-1.429×10^{-4}	[-0.0038, 0.00351]	1.0000	
adaptive	seq2seq	1000	-0.001693	[-0.00535, 0.00196]	0.6739	
adaptive	seq2seq	1e+04	-0.008105	[-0.0118, -0.00445]	$< 10^{-4}$	*
tropical	pointer	0.01	0.002142	[-0.00552, 0.00981]	0.9520	
tropical	pointer	0.1	0.001889	[-0.00578, 0.00955]	0.9743	
tropical	pointer	1	0.001886	[-0.00578, 0.00955]	0.9745	
tropical	pointer	10	0.001439	[-0.00623, 0.0091]	0.9942	
tropical	pointer	100	0.002259	[-0.00541, 0.00992]	0.9385	
tropical	pointer	1000	0.002012	[-0.00565, 0.00968]	0.9646	
tropical	pointer	1e+04	0.002726	[-0.00494, 0.0104]	0.8640	
tropical	seq2seq	0.01	0.001313	[-0.00323, 0.00586]	0.9439	
tropical	seq2seq	0.1	9.355×10^{-4}	[-0.00361, 0.00548]	0.9903	
tropical	seq2seq	1	8.775×10^{-4}	[-0.00367, 0.00542]	0.9932	
tropical	seq2seq	10	7.615×10^{-4}	[-0.00378, 0.00531]	0.9971	
tropical	seq2seq	100	2.869×10^{-5}	[-0.00452, 0.00457]	1.0000	
tropical	seq2seq	1000	-0.00128	[-0.00582, 0.00326]	0.9502	
tropical	seq2seq	1e+04	-0.009454	[-0.014, -0.00491]	$< 10^{-4}$	*
vanilla	pointer	0.01	6.832×10^{-4}	$[-8.87 \times 10^{-4}, 0.00225]$	0.7273	
vanilla	pointer	0.1	4.147×10^{-4}	[-0.00116, 0.00198]	0.9635	
vanilla	pointer	1	3.229×10^{-4}	[-0.00125, 0.00189]	0.9903	
vanilla	pointer	10	7.451×10^{-4}	$[-8.25 \times 10^{-4}, 0.00231]$	0.6509	
vanilla	pointer	100	8.221×10^{-4}	$[-7.48 \times 10^{-4}, 0.00239]$	0.5550	
vanilla	pointer	1000	0.001016	$[-5.54 \times 10^{-4}, 0.00259]$	0.3400	
vanilla	pointer	1e+04	7.178×10^{-4}	$[-8.52 \times 10^{-4}, 0.00229]$	0.6851	
vanilla	seq2seq	0.01	0.001096	[-0.00197, 0.00416]	0.8606	
vanilla	seq2seq	0.1	0.001031	[-0.00203, 0.00409]	0.8906	
vanilla	seq2seq	1	0.001153	[-0.00191, 0.00422]	0.8315	
vanilla	seq2seq	10	0.001044	[-0.00202, 0.00411]	0.8847	
vanilla	seq2seq	100	0.00122	[-0.00184, 0.00428]	0.7946	
vanilla	seq2seq	1000	-0.001033	[-0.0041, 0.00203]	0.8896	
vanilla	seq2seq	1e+04	-0.008597	[-0.0117, -0.00554]	$< 10^{-4}$	*

Table 27: `id_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-1.835×10^{-5}	$[-7.11 \times 10^{-5}, 2.83 \times 10^{-5}]$	0.8997	-0.0286	0.8416
adaptive	seq2seq	-0.001055	$[-0.00119, -8.49 \times 10^{-4}]$	$< 10^{-4}$	-0.459	0.0008
tropical	pointer	8.474×10^{-5}	$[-8.85 \times 10^{-5}, 2.58 \times 10^{-4}]$	0.8066	0.00952	0.9486
tropical	seq2seq	-0.001342	[-0.00148, -0.00121]	$< 10^{-4}$	-0.432	0.0012
vanilla	pointer	6.449×10^{-5}	$[-2.89 \times 10^{-5}, 1.44 \times 10^{-4}]$	0.3991	0.0971	0.4777
vanilla	seq2seq	-0.001184	$[-0.0016, -8.05 \times 10^{-4}]$	$< 10^{-4}$	-0.482	0.0006

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.002761	[-0.003, 0.00853]	0.7368
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.00422	[-0.00125, 0.00969]	0.7368

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.004236	[-0.00121, 0.00968]	0.7368	
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.004325	[-0.00138, 0.01]	0.7368	
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.003456	[-0.00214, 0.00905]	0.7368	
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.004198	[-0.00197, 0.0104]	0.7368	
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.003757	[-0.00283, 0.0103]	0.7368	
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.00477	[-0.00124, 0.0108]	0.7368	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.008669	[0.00542, 0.0119]	0.0284	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.008886	[0.00468, 0.0131]	0.0508	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.008574	[0.00392, 0.0132]	0.0763	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.008394	[0.00523, 0.0116]	0.0284	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.008386	[0.00356, 0.0132]	0.0770	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.007478	[0.00329, 0.0117]	0.0770	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.008423	[0.00493, 0.0119]	0.0336	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.007813	[0.00483, 0.0108]	0.0284	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.004253	[0.00247, 0.00604]	0.0298	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.004941	[0.00282, 0.00706]	0.0298	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.00533	[0.00354, 0.00712]	0.0165	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.00524	[0.00329, 0.00719]	0.0222	*
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.004901	[0.00282, 0.00699]	0.0298	*
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.004479	[0.0028, 0.00616]	0.0222	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.004631	[0.00311, 0.00615]	0.0161	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.004887	[0.00332, 0.00645]	0.0156	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	8.006×10^{-4}	[-0.00109, 0.0027]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-4.780×10^{-5}	[-0.00287, 0.00278]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	2.458×10^{-4}	[-0.00262, 0.00311]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-3.399×10^{-4}	[-0.00328, 0.0026]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-2.161×10^{-4}	[-0.0033, 0.00287]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-5.623×10^{-4}	[-0.00388, 0.00275]	1.0000	

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	1.414×10^{-4}	$[-7.11 \times 10^{-4}, 9.93 \times 10^{-4}]$	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.001293	$[-4.73 \times 10^{-4}, 0.00306]$	0.8954	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.008559	[0.0078, 0.00932]	0.0001	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.009682	[0.00779, 0.0116]	0.0024	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.009574	[0.00754, 0.0116]	0.0029	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.009856	[0.00812, 0.0116]	0.0019	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.009924	[0.00812, 0.0117]	0.0019	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.00975	[0.00801, 0.0115]	0.0019	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.01165	[0.0109, 0.0124]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.01802	[0.0164, 0.0196]	0.0001	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-8.019×10^{-4}	[-0.00707, 0.00547]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	2.710×10^{-5}	[-0.00619, 0.00624]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	1.518×10^{-4}	[-0.00529, 0.00559]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	2.067×10^{-4}	[-0.00573, 0.00614]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-1.240×10^{-4}	[-0.00552, 0.00527]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.001429	[-0.00448, 0.00734]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.00249	[-0.0046, 0.00958]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.01138	[0.00514, 0.0176]	0.0607	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.005106	[0.00357, 0.00664]	0.0107	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.004693	[0.00317, 0.00622]	0.0134	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.00449	[0.00258, 0.0064]	0.0312	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.004276	[0.00272, 0.00583]	0.0189	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.004807	[0.00253, 0.00709]	0.0424	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.004708	[0.00217, 0.00725]	0.0607	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.007156	[0.00617, 0.00814]	0.0007	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.01442	[0.0116, 0.0172]	0.0024	*

A.7 `id_exact`

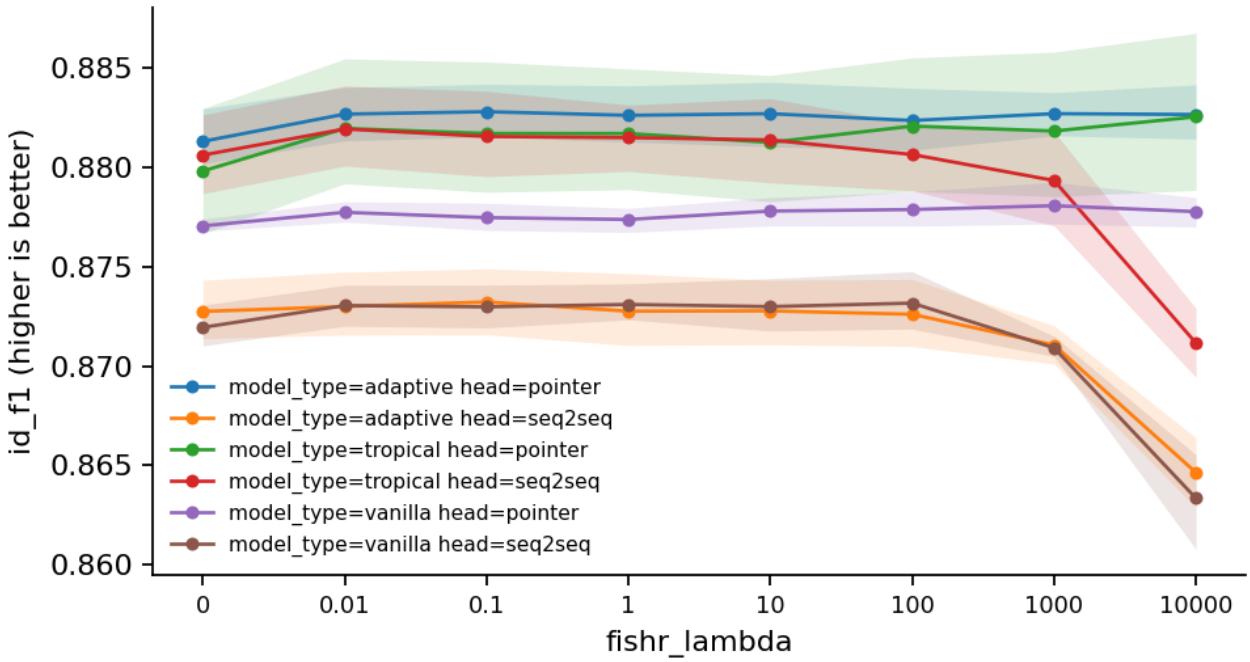


Figure 7: `id_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

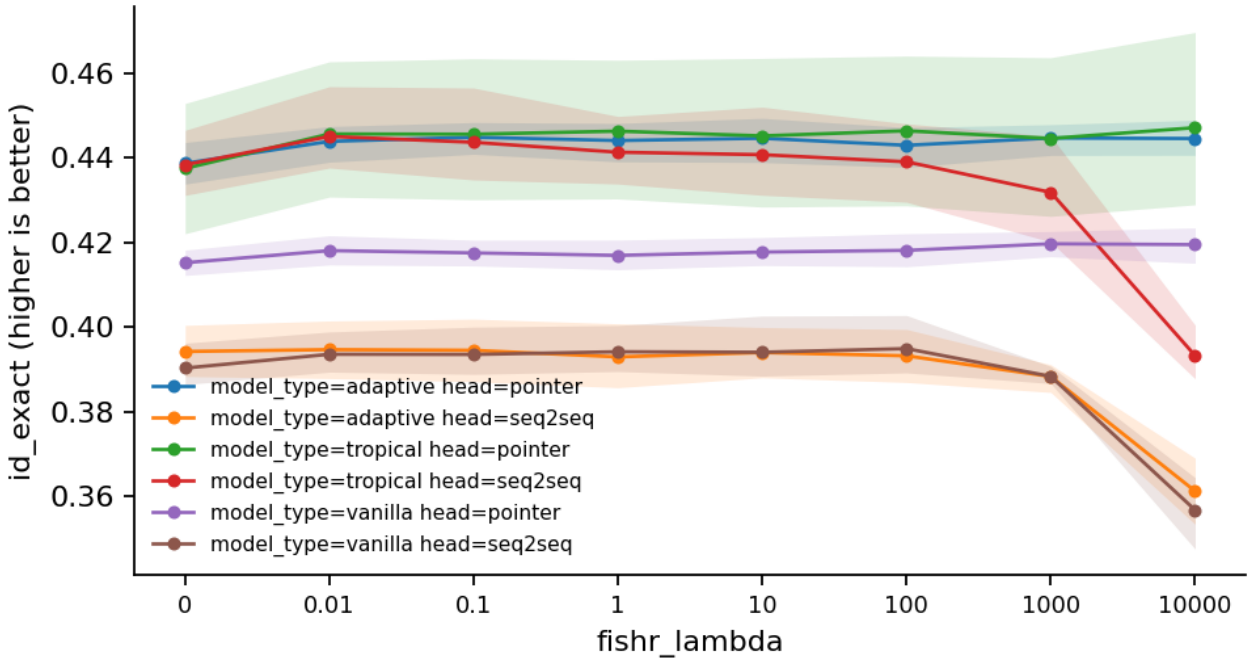


Figure 8: `id_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 29: `id_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.4388	[0.434, 0.444]	0.00652	5	0
adaptive	pointer	0.01	0.4439	[0.439, 0.447]	0.00575	5	0
adaptive	pointer	0.1	0.4449	[0.441, 0.448]	0.00504	5	0
adaptive	pointer	1	0.4441	[0.439, 0.448]	0.00552	5	0
adaptive	pointer	10	0.4447	[0.439, 0.449]	0.00694	5	0
adaptive	pointer	100	0.443	[0.438, 0.447]	0.00597	5	0
adaptive	pointer	1000	0.4447	[0.441, 0.448]	0.00471	5	0
adaptive	pointer	1e+04	0.4446	[0.441, 0.449]	0.0055	5	0
adaptive	seq2seq	0	0.3943	[0.388, 0.4]	0.00811	5	0
adaptive	seq2seq	0.01	0.3947	[0.388, 0.402]	0.00879	5	0
adaptive	seq2seq	0.1	0.3946	[0.387, 0.402]	0.00981	5	0
adaptive	seq2seq	1	0.393	[0.386, 0.401]	0.00965	5	0
adaptive	seq2seq	10	0.394	[0.388, 0.4]	0.00788	5	0
adaptive	seq2seq	100	0.3933	[0.387, 0.4]	0.00806	5	0
adaptive	seq2seq	1000	0.3884	[0.385, 0.391]	0.00416	5	0
adaptive	seq2seq	1e+04	0.3613	[0.353, 0.369]	0.0101	5	0
tropical	pointer	0	0.4375	[0.422, 0.453]	0.0209	5	0
tropical	pointer	0.01	0.4457	[0.431, 0.463]	0.0212	5	0
tropical	pointer	0.1	0.4456	[0.43, 0.463]	0.0211	5	0
tropical	pointer	1	0.4464	[0.43, 0.463]	0.0226	5	0
tropical	pointer	10	0.4452	[0.428, 0.464]	0.0223	5	0
tropical	pointer	100	0.4464	[0.429, 0.464]	0.0243	5	0
tropical	pointer	1000	0.4446	[0.426, 0.464]	0.025	5	0
tropical	pointer	1e+04	0.4472	[0.429, 0.47]	0.0256	5	0
tropical	seq2seq	0	0.4384	[0.431, 0.447]	0.0104	5	0
tropical	seq2seq	0.01	0.4451	[0.438, 0.457]	0.013	5	0
tropical	seq2seq	0.1	0.4437	[0.435, 0.457]	0.0141	5	0
tropical	seq2seq	1	0.4414	[0.434, 0.45]	0.0101	5	0
tropical	seq2seq	10	0.4408	[0.431, 0.452]	0.0132	5	0
tropical	seq2seq	100	0.4391	[0.43, 0.448]	0.0115	5	0
tropical	seq2seq	1000	0.4319	[0.42, 0.445]	0.016	5	0
tropical	seq2seq	1e+04	0.3933	[0.388, 0.401]	0.00817	5	0
vanilla	pointer	0	0.4153	[0.412, 0.418]	0.00408	5	0
vanilla	pointer	0.01	0.4181	[0.415, 0.422]	0.00441	5	0
vanilla	pointer	0.1	0.4176	[0.415, 0.421]	0.00408	5	0
vanilla	pointer	1	0.417	[0.414, 0.421]	0.00469	5	0
vanilla	pointer	10	0.4178	[0.415, 0.421]	0.00419	5	0
vanilla	pointer	100	0.4182	[0.414, 0.422]	0.0051	5	0
vanilla	pointer	1000	0.4197	[0.417, 0.423]	0.0038	5	0
vanilla	pointer	1e+04	0.4195	[0.415, 0.424]	0.00559	5	0
vanilla	seq2seq	0	0.3904	[0.387, 0.396]	0.00657	5	0
vanilla	seq2seq	0.01	0.3936	[0.389, 0.399]	0.00605	5	0
vanilla	seq2seq	0.1	0.3936	[0.389, 0.4]	0.0072	5	0
vanilla	seq2seq	1	0.3942	[0.39, 0.4]	0.00708	5	0
vanilla	seq2seq	10	0.3941	[0.388, 0.403]	0.00958	5	0
vanilla	seq2seq	100	0.395	[0.389, 0.403]	0.00883	5	0
vanilla	seq2seq	1000	0.3884	[0.387, 0.391]	0.00273	5	0
vanilla	seq2seq	1e+04	0.3568	[0.348, 0.364]	0.011	5	0

Table 30: `id_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	0.00518	[-0.00492, 0.0153]	0.5764	
adaptive	pointer	0.1	0.00616	[-0.00394, 0.0163]	0.3995	
adaptive	pointer	1	0.00538	[-0.00472, 0.0155]	0.5382	
adaptive	pointer	10	0.00592	[-0.00418, 0.016]	0.4400	
adaptive	pointer	100	0.00426	[-0.00584, 0.0144]	0.7527	
adaptive	pointer	1000	0.00598	[-0.00412, 0.0161]	0.4297	
adaptive	pointer	1e+04	0.00586	[-0.00424, 0.016]	0.4505	
adaptive	seq2seq	0.01	4.400×10^{-4}	[-0.0144, 0.0153]	1.0000	
adaptive	seq2seq	0.1	2.800×10^{-4}	[-0.0146, 0.0151]	1.0000	
adaptive	seq2seq	1	-0.00128	[-0.0161, 0.0136]	1.0000	
adaptive	seq2seq	10	-2.400×10^{-4}	[-0.0151, 0.0146]	1.0000	
adaptive	seq2seq	100	-0.00102	[-0.0159, 0.0138]	1.0000	
adaptive	seq2seq	1000	-0.00592	[-0.0208, 0.00894]	0.7950	
adaptive	seq2seq	1e+04	-0.033	[-0.0479, -0.0181]	$< 10^{-4}$	*
tropical	pointer	0.01	0.00816	[-0.0319, 0.0482]	0.9908	
tropical	pointer	0.1	0.0081	[-0.032, 0.0482]	0.9912	
tropical	pointer	1	0.00882	[-0.0312, 0.0489]	0.9859	
tropical	pointer	10	0.0077	[-0.0324, 0.0478]	0.9934	
tropical	pointer	100	0.0089	[-0.0312, 0.049]	0.9852	
tropical	pointer	1000	0.0071	[-0.033, 0.0472]	0.9959	
tropical	pointer	1e+04	0.00964	[-0.0304, 0.0497]	0.9773	
tropical	seq2seq	0.01	0.0067	[-0.0148, 0.0282]	0.9207	
tropical	seq2seq	0.1	0.0053	[-0.0162, 0.0268]	0.9741	
tropical	seq2seq	1	0.00296	[-0.0185, 0.0244]	0.9991	
tropical	seq2seq	10	0.00238	[-0.0191, 0.0239]	0.9998	
tropical	seq2seq	100	7.000×10^{-4}	[-0.0208, 0.0222]	1.0000	
tropical	seq2seq	1000	-0.00652	[-0.028, 0.015]	0.9297	
tropical	seq2seq	1e+04	-0.0451	[-0.0666, -0.0236]	$< 10^{-4}$	*
vanilla	pointer	0.01	0.00286	[-0.00505, 0.0108]	0.8550	
vanilla	pointer	0.1	0.00232	[-0.00559, 0.0102]	0.9398	
vanilla	pointer	1	0.00174	[-0.00617, 0.00965]	0.9859	
vanilla	pointer	10	0.00252	[-0.00539, 0.0104]	0.9131	
vanilla	pointer	100	0.00292	[-0.00499, 0.0108]	0.8432	
vanilla	pointer	1000	0.00448	[-0.00343, 0.0124]	0.4747	
vanilla	pointer	1e+04	0.00428	[-0.00363, 0.0122]	0.5217	
vanilla	seq2seq	0.01	0.00326	[-0.0103, 0.0168]	0.9772	
vanilla	seq2seq	0.1	0.00322	[-0.0103, 0.0168]	0.9787	
vanilla	seq2seq	1	0.00386	[-0.00968, 0.0174]	0.9474	
vanilla	seq2seq	10	0.00374	[-0.0098, 0.0173]	0.9547	
vanilla	seq2seq	100	0.0046	[-0.00894, 0.0181]	0.8866	
vanilla	seq2seq	1000	-0.00196	[-0.0155, 0.0116]	0.9988	
vanilla	seq2seq	1e+04	-0.03354	[-0.0471, -0.02]	$< 10^{-4}$	*

Table 31: `id_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	2.000×10^{-5}	$[-1.95 \times 10^{-4}, 2.35 \times 10^{-4}]$	0.9644	-0.0171	0.9072
adaptive	seq2seq	-0.004016	[-0.00464, -0.00338]	0.0002	-0.385	0.0020
tropical	pointer	9.000×10^{-5}	$[-7.22 \times 10^{-4}, 9.02 \times 10^{-4}]$	0.9606	0.0133	0.9354
tropical	seq2seq	-0.006475	[-0.00712, -0.00583]	$< 10^{-4}$	-0.436	0.0008
vanilla	pointer	3.486×10^{-4}	$[-4.07 \times 10^{-5}, 7.23 \times 10^{-4}]$	0.3415	0.105	0.4429

Table 31: `id_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
vanilla	seq2seq	-0.004286	[-0.00571, -0.00291]	0.0002	-0.444	0.0010

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.02228	[-0.00221, 0.0468]	0.2778	
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.02758	[0.00322, 0.0519]	0.2778	
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.02806	[0.0029, 0.0532]	0.2778	
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.02936	[0.00305, 0.0557]	0.2778	
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.02746	$[8.05 \times 10^{-4}$, 0.0541]	0.2778	
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.02826	$[-1.26 \times 10^{-4}$, 0.0566]	0.2778	
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.0249	[-0.00505, 0.0548]	0.2778	
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.02764	$[9.13 \times 10^{-4}$, 0.0544]	0.2778	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.04806	[0.0325, 0.0637]	0.0164	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.0515	[0.0314, 0.0716]	0.0267	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.05014	[0.0275, 0.0727]	0.0423	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.04716	[0.03, 0.0643]	0.0228	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.0467	[0.0249, 0.0685]	0.0423	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.04416	[0.0237, 0.0646]	0.0423	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.0435	[0.0237, 0.0633]	0.0423	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.0365	[0.0234, 0.0496]	0.0228	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.0235	[0.0133, 0.0337]	0.0273	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.02582	[0.0166, 0.0351]	0.0194	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.02734	[0.0188, 0.0359]	0.0135	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.02714	[0.0173, 0.037]	0.0194	*
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.0269	[0.0172, 0.0366]	0.0194	*
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.02484	[0.017, 0.0326]	0.0135	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.025	[0.0179, 0.0321]	0.0099	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.02508	[0.0154, 0.0347]	0.0197	*

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.00392	[-0.00756, 0.0154]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.0011	[-0.0112, 0.0134]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	9.800×10^{-4}	[-0.0125, 0.0144]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.00122	[-0.0162, 0.0137]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-6.000×10^{-5}	[-0.0133, 0.0132]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.0017	[-0.0142, 0.0108]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-4.000×10^{-5}	[-0.00464, 0.00456]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.00446	[-0.0055, 0.0144]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.04448	[0.0398, 0.0491]	0.0003	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.04922	[0.0403, 0.0581]	0.0018	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.05036	[0.0411, 0.0596]	0.0018	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.05114	[0.0448, 0.0575]	0.0005	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.05064	[0.0429, 0.0584]	0.0010	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.04976	[0.0438, 0.0557]	0.0004	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.05638	[0.0523, 0.0605]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.08334	[0.0759, 0.0908]	0.0001	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-8.800×10^{-4}	[-0.0284, 0.0267]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	5.800×10^{-4}	[-0.0279, 0.0291]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.00192	[-0.0245, 0.0283]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.00498	[-0.0198, 0.0298]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.00444	[-0.0227, 0.0316]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.00732	[-0.0226, 0.0372]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.01274	[-0.0254, 0.0509]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.05386	[0.0244, 0.0833]	0.0570	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.0249	[0.0195, 0.0303]	0.0030	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.0245	[0.0184, 0.0306]	0.0045	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.024	[0.0165, 0.0315]	0.0097	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.02278	[0.0176, 0.028]	0.0034	*

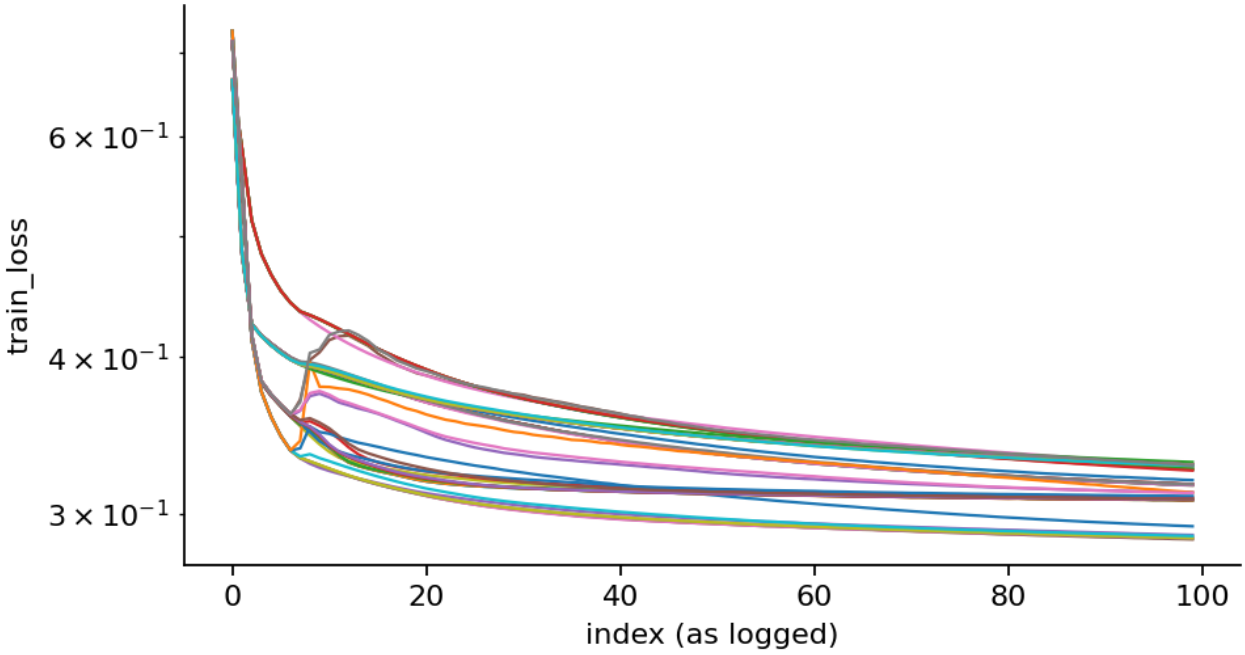


Figure 9: `train_loss`, the mean over seeds per cell.

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.02368	[0.0144, 0.033]	0.0209	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.02322	[0.0134, 0.0331]	0.0252	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.03134	[0.0266, 0.0361]	0.0010	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.06272	[0.0512, 0.0742]	0.0018	*

B Curves

C Configuration

The base configuration every run shares (frozen hash `057e46d824d4`); factor levels are in Table 1.

```
task:
  name: KnapsackDataset
  value_range:
  - 1
  - 10
  target_range:
  - 10
  - 20
  weight_range:
  - 1
  - 10
  adversarial_range:
  - 10
```

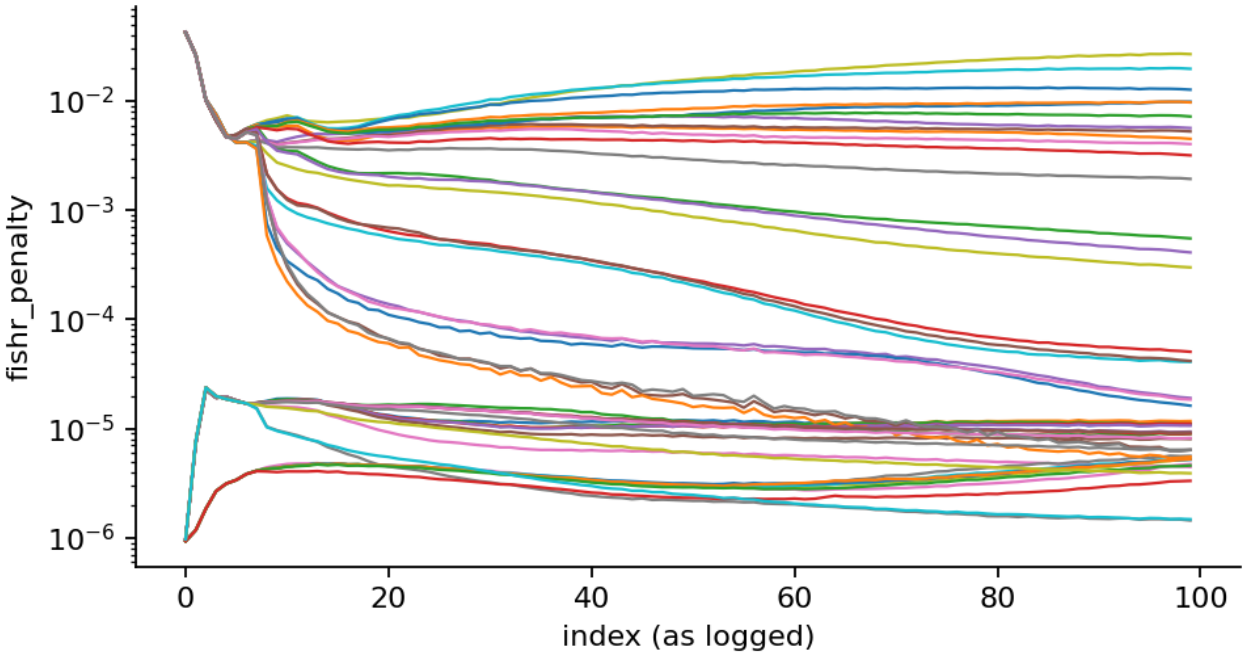


Figure 10: `fishr_penalty`, the mean over seeds per cell.

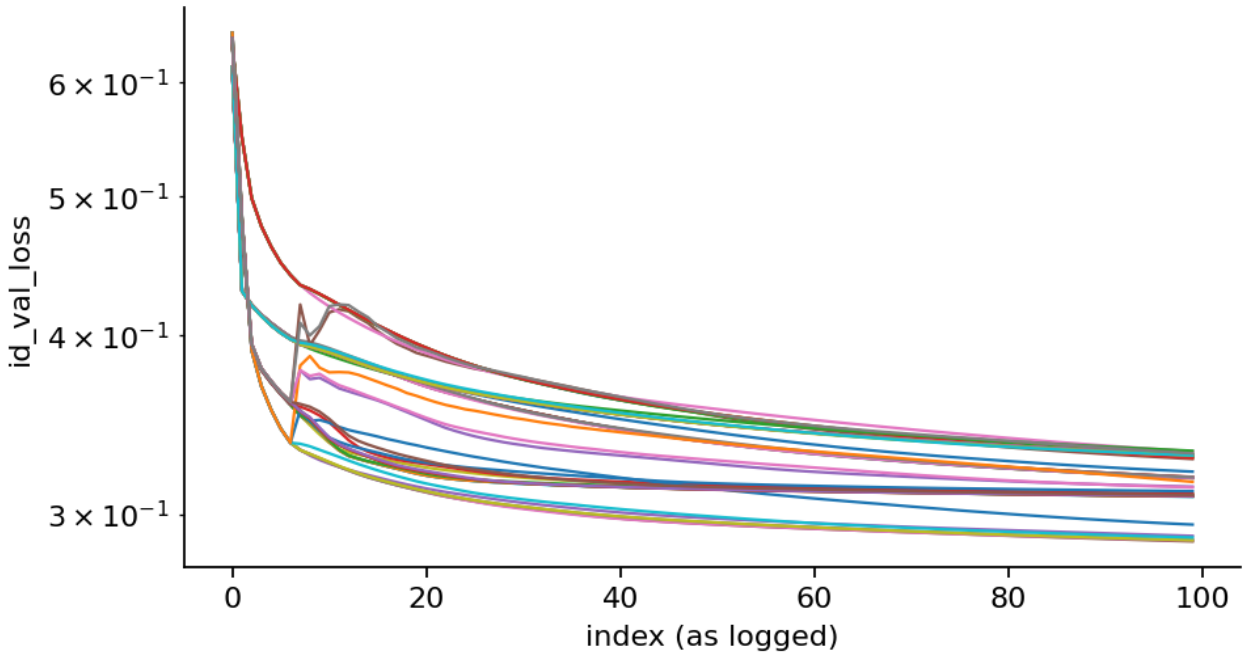


Figure 11: `id_val_loss`, the mean over seeds per cell.

```
  - 30
  noise_prob: 0.0
data:
  train_lengths:
  - 4
  - 5
  - 6
  - 7
  - 8
  ood_lengths:
  - 12
  - 16
  - 64
  ood_value_range:
  - 11
  - 21
  ood_noise_range:
  - 10
  - 30
  ood_noise_prob: 0.5
  n_per_length: 20000
  n_eval_per_length: 2000
  curve_eval_limit: 256
model:
  d_model: 64
  n_heads: 2
  num_layers: 1
  dropout: 0.0
  tropical_impl: legacy
optim:
  lr: 0.0001
  epochs: 100
  batch_size: 100
  eval_batch_size: 256
fishr:
  ema: 0.95
  anneal_iters: 1500
  reset_optimizer: true
```

Store: `campaigns/tropical/Knapsack/results`.